# Frequentist and Bayesian Analysis of the Matter–Antimatter Asymmetry in $B^{\pm} \to K^{\pm}K^{+}K^{-}$ Decays at LHCb

**Final project for the *Statistics and Data Analysis* course**
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We measure the raw charge asymmetry $A_{\rm raw} = [N(B^-) - N(B^+)]/[N(B^-) + N(B^+)]$ of $B^{\pm} \to K^{\pm}K^{+}K^{-}$ decays in the 2011 LHCb Open Data record with an extended unbinned maximum-likelihood fit and with a Bayesian analysis of the same likelihood. On 19,731 candidates we find $A_{\rm raw} = -0.0441 \pm 0.0085\,(\text{stat}) \pm 0.0072\,(\text{syst})$ (Frequentist) and $-0.0443 \pm 0.0085 \pm 0.0070$ (Bayesian), 5.2 standard deviations from zero. The methods agree in value, uncertainty and coverage, as verified on toy-MC experiments, and the dominant systematic uncertainty is the particle-identification requirement. In the Dalitz region of the published analysis we find $-0.234 \pm 0.025$, reproducing $-0.239 \pm 0.020$. $A_{\rm raw}$ is not corrected for production and detection asymmetries and is not a measurement of $A_{CP}$.

## I. INTRODUCTION

The observable of this work is the raw charge asymmetry of the signal yields of $B^{\pm} \to K^{\pm}K^{+}K^{-}$ decays,

$$A_{\rm raw} = \frac{N(B^-) - N(B^+)}{N(B^-) + N(B^+)}, \tag{1}$$

which in three-body charmless $B$ decays can be studied globally and locally over the Dalitz plane [1, 2]. It differs from the $CP$ asymmetry $A_{CP}$ by the production asymmetry $A_P$ of $B^{\pm}$ in $pp$ collisions and by the detection asymmetry $A_D$ of the final state [1],

$$A_{CP} \approx A_{\rm raw} - A_P - A_D. \tag{2}$$

Neither correction is applied here, so $A_{\rm raw}$ is not a measurement of $A_{CP}$; the corrections of the published analysis are discussed in Sec. IX.

We use the LHCb Open Data record for this decay [3, 4], a subset of the 2011 data on which the first evidence of an inclusive $CP$ asymmetry was published [5], to ask: *to what extent do Frequentist likelihood fits and Bayesian inference produce consistent, well-controlled estimates of* $A_{\rm raw}$*?* The aim is a methodological comparison of the two frameworks on the same data and on the same likelihood, so that any difference comes from the inference and not from the model; the record lacks the control channel needed for $A_P$ and $A_D$ (Appendix J.5). Secs. II–III describe the data and the common model, Secs. IV–V the two fits, and Secs. VI–IX the systematic study, the validation, the Dalitz plane and the comparison, which includes a machine-learning selection (Appendix I). The main results are in Table I.

## II. DATA

*In short: the record, the common preselection (*23,895 *candidates) and the fit window (*19,731 *candidates); the spectrum peaks at* 5284 *MeV for both charges and both polarities, and the counting asymmetry is* $-0.043 \pm 0.007$.

### A. Dataset, Preselection and Mass

The dataset is the CERN Open Data record 4900 [3]: $B^{\pm} \to h^{\pm}h^{+}h^{-}$ candidates collected by LHCb in 2011 at $\sqrt{s} = 7\,\text{TeV}$, already through trigger and offline selection, split by magnet polarity (8,556,118 candidates), and a phase-space simulation of $B^{\pm} \to K^{\pm}K^{+}K^{-}$ (50,000 decays, no asymmetry) used only to test the local-asymmetry procedure (Sec. VIII). The same preselection is used in all analyses,

$$\texttt{H*_ProbPi} < 0.5 \ \wedge\ \texttt{H*_ProbK} > 0.5 \ \wedge\ \neg\texttt{H*_isMuon},$$

for each of the three tracks, and retains 23,895 candidates (0.28%) (Appendix A). $H1$ is always the kaon of charge opposite to the $B$, so the Dalitz variables are $m^2(H1, H2)$ and $m^2(H1, H3)$.

The three tracks are assigned the charged-kaon mass and their four-momenta are summed to reconstruct $m(KKK)$. The nominal fit window, common to the two analyses, is $[5100, 5600]$ MeV and contains 19,731 candidates (11,636 magnet down, 8095 magnet up). Below ∼5150 MeV the background rises, which a single exponential cannot describe; the window includes the start of this rise, described by a second exponential in the background model (Secs. IV–V).

### B. Data Exploration

Figure I shows the spectrum in the fit window. The peak at $m \approx 5284\,\text{MeV}$ over a falling combinatorial background has the same shape for the two charges and the two polarities, which allows a mass and a resolution common to $B^+$ and $B^-$ in the model (Sec. III). The

| Method | Sample | $A_{\rm raw}$ | Statistical | Systematic | Section |
|---|---|---|---|---|---|
| Counting (no fit) | fit window, 19,731 | $-0.0432$ | 0.0071 | – | II B |
| Frequentist fit | fit window, 19,731 | $-0.0441$ | 0.0085 | 0.0072 | IV, VI |
| Bayesian fit | fit window, 19,731 | $-0.0443$ | 0.0085 | 0.0070 | V, VI |
| Bayesian fit, $D^0$ veto | 18,744 | $-0.0496$ | 0.0088 | – | App. D.15 |
| Bayesian fit, ML selection | score $> 0.5$, 15,709 | $-0.0451$ | 0.0087 | – | App. I.9 |
| Frequentist fit, ML selection | score $> 0.5$, 15,709 | $-0.0447$ | 0.0086 | – | App. I.9 |
| Bayesian fit, Dalitz region of Ref. [5] | 2146 | $-0.234$ | 0.025 | – | VIII |
| Published analysis (same data) [5] | different selection and model | $-0.060$ | 0.007 | – | – |

TABLE I. Summary of the raw asymmetry $A_{\rm raw}$ (not corrected for production and detection asymmetries). The Bayesian value is the posterior median and its standard deviation; the systematic uncertainty is given only for the nominal fits (Sec. VI). The published $A_{\rm raw}$ comes from a different selection and a model with additional background components, so it is not directly comparable with the first rows.

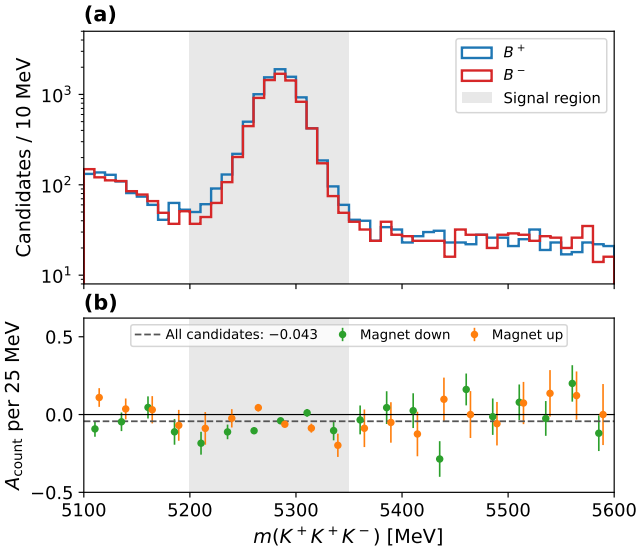



FIG. I. (a) Reconstructed $B \to KKK$ invariant mass in the nominal fit window $[5100, 5600]$ MeV for $B^+$ and $B^-$, logarithmic scale; the signal region $[5200, 5350]$ MeV is shaded. (b) Counting asymmetry in 25 MeV bins of the mass for the two magnet polarities, and for all candidates (dashed); the two polarities agree within their uncertainties and the asymmetry is similar in the peak and in the sidebands.

counting asymmetry of all candidates, signal and background together, is $A_{\rm count} = (N^- - N^+)/(N^- + N^+) = -0.043 \pm 0.007$ (binomial uncertainty); the fits of Secs. IV and V refine it by separating signal and background.

### C. Additional Results

*Branches and cut flow.* The analysis reads momentum, particle-identification likelihoods, muon flag and charge of each track; the kaon-identification requirement on the three tracks is by far the most selective (Appendix A.3).

*Charge combinations.* $H1$ is always the kaon of charge opposite to the $B$ and $H2$, $H3$ the same-sign pair, so the Dalitz variables are the two opposite-sign invariant masses (Appendix A).

*Shared tracks and window.* Only 42 candidates of the fit window (0.2%) share a track with another candidate and all are kept; the mass is computed with the kaon hypothesis on all tracks, and the window includes the start of the rise of the background below $\sim$ 5150 MeV (Appendix A).

## III. MASS MODEL AND LIKELIHOOD

*In short: one extended likelihood shared by the two analyses: a Gaussian signal with common mean and width, a double-exponential background for each charge and $A$ as a parameter.* The mass spectrum of each charge sample is described by a signal-plus-background mixture on the nominal fit window $[5100, 5600]$ MeV (Sec. II A),

$$f^{\pm}(m) = N_s^{\pm}\, S(m; \mu, \sigma) + N_b^{\pm}\, B^{\pm}(m), \qquad (3)$$

where $S$ is a Gaussian truncated to the window, with mean $\mu$ and width $\sigma$ shared by the two charges, and the background is a double exponential with a shape of its own for each charge,

$$B^{\pm}(m) = f_{\pm}\, E(m; \lambda_1^{\pm}) + (1 - f_{\pm})\, E(m; \lambda_2^{\pm}),$$

with $E(m;\lambda) \propto e^{-\lambda m}$ a truncated exponential normalized on the window. The steep component ($\lambda_1$) describes the rise of the background towards the lower edge of the window, the slow one ($\lambda_2$) the combinatorial background under and above the peak. The ordering $\lambda_1 > \lambda_2$ removes the label-switching symmetry of the mixture. Signal and background yields are split between the charges as

$$N_s^{\pm} = \tfrac{N_s}{2}(1 \mp A), \qquad N_b^{\pm} = \tfrac{N_b}{2}(1 \mp A_b),$$

so that $A = (N_s^- - N_s^+)/(N_s^- + N_s^+)$, the raw asymmetry $A_{\rm raw}$ of Eq. (1), is itself a parameter of the fit, the parameter of interest (POI, all the others being nuisance parameters), and the background asymmetry $A_b$ is free. The parameter vector has twelve components, $\theta = (N_s, N_b, A, A_b, \mu, \sigma, \lambda_1^{\pm}, \lambda_2^{\pm}, f_{\pm})$.

Given the data $D = \{(m_i, q_i)\}_{i=1}^{N}$ (mass and charge of each candidate), the extended unbinned likelihood of the two charge samples is

$$\mathcal{L}(D \mid \theta) \propto e^{-(N_s+N_b)} \prod_{i:\, q_i=+1} \left[N_s^+ S(m_i) + N_b^+ B^+(m_i)\right] \times \prod_{i:\, q_i=-1} \left[N_s^- S(m_i) + N_b^- B^-(m_i)\right], \quad (4)$$

where $S(m_i) \equiv S(m_i; \mu, \sigma)$. The likelihood is extended (the yields are parameters, hence the Poisson term (Appendix B)); it is the common ground of the two mass analyses: Sec. IV maximizes it and Sec. V multiplies it by a prior. Since both use this model unchanged, any difference between their results can only come from the inference.

### A. Additional Results

*Minimal model.* A six-parameter model on a narrower window gives $A_{\text{raw}} = -0.0497$ (68% CI $[-0.058, -0.041]$), compatible with the nominal result within its uncertainty; it is the model of the low-statistics comparison (Appendix B.4).

## IV. FREQUENTIST ANALYSIS

*In short: the likelihood maximized for the two charges together gives* $N_s = 15{,}328 \pm 133$ *and* $\hat{A} = -0.0441$ *with Minos intervals;* $A = 0$ *is rejected at* $5.2\sigma$ *and the interval has nominal coverage in* 1000 *bootstrap replicas.* The Frequentist analysis estimates $A_{\text{raw}}$ by maximizing the extended unbinned likelihood of Eq. (4) with `iminuit` [6]; the uncertainty and the test of $A = 0$ come from the profile likelihood, the other parameters being nuisance parameters (Secs. IV A–IV B).

### A. Mass Model and Fit Strategy

The common mass model of Sec. III (Eq. (3)) was fitted on $[5100, 5600]$ MeV: a Gaussian signal with mean $\mu$ and width $\sigma$ shared by the two charges and a double-exponential background with shape and yield of its own for each charge, twelve parameters as in the Bayesian fit, through `iminuit.cost.ExtendedUnbinnedNLL`; the minimization started from an unbinned fit of the sidebands (the window without $[5200, 5350]$ MeV). The asymmetry was estimated in two ways: in the *simultaneous* fit the two charges share $N_s^{\pm} = N_s(1 \mp A)/2$ and $A$ is a parameter of the likelihood; in the *separate* fits each charge was fitted alone (seven free parameters each) and $A_{\text{raw}} = (N_s^- - N_s^+)/(N_s^- + N_s^+)$ was computed from the two yields. They give $-0.0441 \pm 0.0085$ and $-0.0442 \pm 0.0087$: sharing $\mu$ and $\sigma$ between a particle and its antiparticle is adequate, and the simultaneous fit, which gives the profile-likelihood interval directly, is the nominal one. It gives $N_s = 15{,}328 \pm 133$ and $N_b = 4{,}403 \pm 82$, against the Bayesian posterior medians $15{,}328 \pm 133$ and $4404 \pm 82$ (Sec. V C). The shapes were not tuned on this fit, and the alternatives are systematic variants (Sec. VI). The nominal fit has $\chi^2/\text{ndof} = 160/88$ ($p = 4.8 \cdot 10^{-6}$) in $2 \times 50$ bins of 10 MeV; a single exponential is rejected (294/92, $p = 5.7 \cdot 10^{-23}$) (Appendix F.5) because the background rises towards the lower edge of the window (Sec. II A). The fit of the two charges is shown in Fig. II.

### B. Profile Likelihood and Confidence Interval

The maximum-likelihood estimate is $\hat{A} = -0.0441$ (Hesse uncertainty 0.0085), with `MINOS` [7] intervals 68% CI $[-0.0526, -0.0356]$ and 95% CI $[-0.0607, -0.0275]$; the uncertainties are statistical only (the systematic one is in Sec. VI). The profile likelihood ratio $q(A) = -2\log[L(A, \hat{\hat{\theta}})/L(\hat{A}, \hat{\theta})]$, with the other eleven parameters re-minimized at each $A$ (`mnprofile`), is shown in Fig. III; by Wilks' theorem [8] it follows a $\chi^2$ distribution with one degree of freedom at the true value of $A$, so the intervals are the sets $q \leq 1$ and $q \leq 3.84$. For the null hypothesis, $q(0) \simeq 27$, $p = 2.05 \times 10^{-7}$, $5.20\sigma$ (Appendix C.1): no asymmetry is rejected, the counterpart of the Savage-Dickey test of Sec. V D ($A = 0$ at 5.2 posterior standard deviations); the $p$-value is not the probability of $H_0$ and the Bayes factor depends on the prior width, but the conclusion is the same. The result agrees with the Bayesian one (Sec. V C: $-0.0443$, 68% CI $[-0.0528, -0.0358]$), as expected with flat priors and $\sim 2 \times 10^4$ events.

### C. Coverage of the Wilks Intervals

Wilks' theorem is asymptotic, so the coverage was checked on the data with 1000 bootstrap replicas of the 19,731 candidates, refitting the nominal model on each (non-converged fits discarded); with $A_{\text{true}} = -0.0441$, the 68% interval of a replica contains it if and only if $q(A_{\text{true}}) \leq 1$. The fraction is $68.4 \pm 1.5\%$ against 68.3% nominal (Fig. IV). It agrees with the toy-MC check of the Bayesian interval ($65.0 \pm 3.3\%$, 200 experiments) (Appendix D.16); the two checks are not equivalent, since the Bayesian pseudo-datasets come from the fitted model (a closure test) while the replicas are resampled from the data and carry the features that the model does not describe, such as the signal tails (Appendix D.13).

### D. Additional Results

*Neyman construction and momentum.* Inverting the test of $A_{\text{true}}$ with toy-MC experiments gives the same

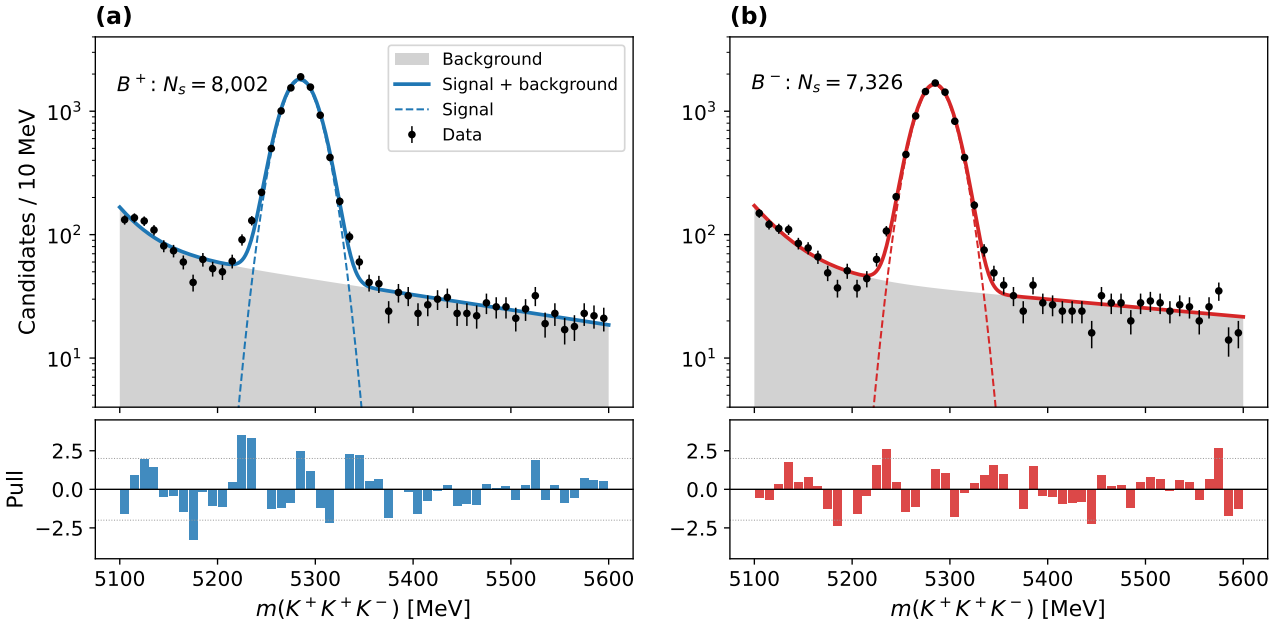



FIG. II. Simultaneous fit of (a) $B^+$ and (b) $B^-$ in 10 MeV bins, logarithmic scale: data, signal (dashed), double-exponential background (grey) and their sum; below, the pulls $(n_i - \mu_i)/\sqrt{\mu_i}$ of the fit, with the $\pm 2$ lines. The fitted signal yields are 8002 and 7326.

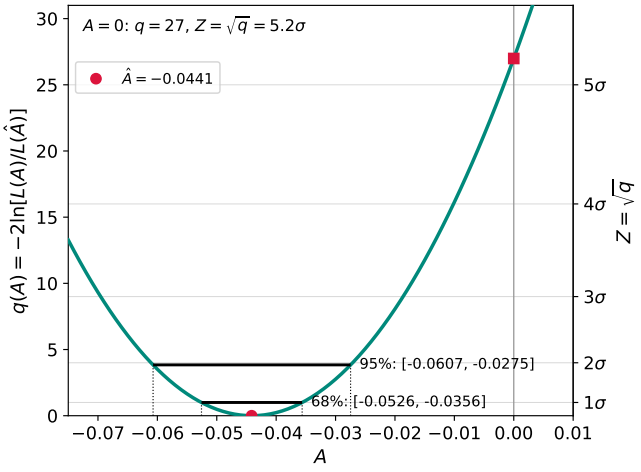



FIG. III. Profile likelihood ratio $q(A)$ of the simultaneous fit, with the other eleven parameters re-minimized at each $A$ (right axis: $Z = \sqrt{q}$). The horizontal segments are the 68% and 95% Minos intervals ($q \leq 1$, $q \leq 3.84$); at $A = 0$, $q \simeq 27$, i.e. $5.2\sigma$.

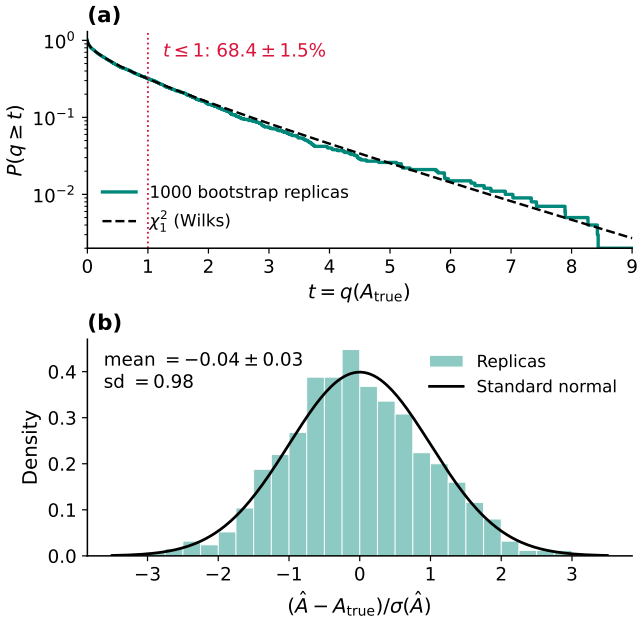



FIG. IV. Coverage check with 1000 bootstrap replicas of the data. (a) Survival function of $t = q(A_{\text{true}})$ against the $\chi^2_1$ distribution expected from Wilks' theorem; the dotted line is the 68.3% limit ($t = 1$). (b) Pulls $(\hat{A} - A_{\text{true}})/\sigma(\hat{A})$ of the replicas against a standard normal.

interval as the likelihood ratio, and $A_{\text{raw}}$ does not depend on the momentum of the $B$ (Appendix C.2).

*Uncertainties of the twelve parameters.* The standard deviations from the Hesse matrix, the bootstrap of the data and the posterior agree for $A$, $A_b$, $\mu$ and $\sigma$; the bootstrap differs for the two yields, because it keeps the total number of candidates fixed, and for the weakly constrained steep slope $\lambda_1^+$ (Appendix C.3).

*Starting values.* 29 of 30 fits from random starting points reach the same minimum (Appendix C.4).

## V. BAYESIAN ANALYSIS

*In short: the same likelihood with uniform priors, sampled with MCMC: $A_{\rm raw} = -0.0443$ (68% interval $[-0.0528, -0.0358]$) with converged chains; the Bayes factor of $A = 0$, $BF_{10} \approx 9 \times 10^3$, depends on the prior width.* The Bayesian analysis takes the likelihood of Sec. III as it is and adds only a prior and a sampler; it changes the question asked of the data, not the model.

### A. Priors and Posterior

Bayes' theorem combines the likelihood of Eq. (4) (Sec. III) with a prior $\pi(\theta)$ into the posterior

$$p(\theta \mid D) = \frac{\mathcal{L}(D \mid \theta)\, \pi(\theta)}{p(D)}, \tag{5}$$

The priors are uniform, with independent ranges except for the ordering of the background slopes: $N_s, N_b \in (0, 5N)$ with $N = 19{,}731$, $A, A_b \in (-1, 1)$, $\mu \in (5260, 5310)$ MeV, $\sigma \in (3, 40)$ MeV, $\lambda_1^{\pm} \in (10^{-5}, 0.1)$ MeV$^{-1}$, $\lambda_2^{\pm} \in (10^{-5}, \lambda_1^{\pm})$, $f_{\pm} \in (0, 1)$. The prior density is therefore a constant on the allowed region $\Theta$,

$$\pi(\theta) \propto \mathbf{1}\,[\theta \in \Theta]\,, \tag{6}$$

and inside it the posterior is proportional to the likelihood. With $\sim 2 \times 10^4$ events the posterior is dominated by the likelihood, as a prior-sensitivity check (Appendix D.15) confirms for $A$. The evidence $p(D)$ is a constant: the MCMC samples $\mathcal{L}\,\pi$ without it, and the Savage-Dickey ratio of Sec. V D avoids it as well; the marginal posterior of $A$ is the histogram of the $A$ component of the chain.

### B. MCMC Sampling and Convergence Diagnostics

The posterior was sampled with the affine-invariant ensemble sampler of Goodman and Weare [9] as implemented in `emcee` [10] (Appendix D.1): 32 walkers started around the maximum of the posterior, 25,000 steps with the first 1/6 discarded as burn-in ($6.7 \times 10^5$ samples, acceptance 0.38). The integrated autocorrelation time is $\tau \in [147, 183]$ steps, so the chain is $N/\tau \geq 113$ autocorrelation times long for every parameter, above the threshold of 50 [10] ($\gtrsim 3600$ effectively independent samples). Two runs started uniformly over the whole prior box converge to the same single mode ($A_{\rm raw} = -0.0446$ and $-0.0442$ (Appendix D.5)).

### C. Posterior Results and Credible Intervals

The nominal model with the posterior medians coincides with the fit of Fig. II: it follows the spectrum over two orders of magnitude, and the pulls show signal tails heavier than a Gaussian, whose effect on $A_{\rm raw}$ is negligible (Appendix D.13). The marginal posterior on the raw asymmetry gives (median and equal-tailed credible intervals, CI)

$$\begin{aligned} A_{\rm raw} &= -0.0443, \\ 68\%\ \text{CI}\ &[-0.0528, -0.0358], \\ 95\%\ \text{CI}\ &[-0.0608, -0.0276]. \end{aligned}$$

The posterior is very close to Gaussian (standard deviation 0.0085), so equal-tailed and highest-posterior-density intervals coincide; with uniform priors its mode, $-0.0441$, is the maximum-likelihood estimate and the result can be compared directly with the Frequentist fit (Sec. IV (Appendix C.1)). The other physics parameters are $N_s = 15{,}328 \pm 133$, $N_b = 4404 \pm 82$, $A_b = -0.040 \pm 0.018$, $\mu = 5284.39 \pm 0.16$ MeV and $\sigma = 18.00 \pm 0.14$ MeV. $A$ is almost uncorrelated with them, which is why profiling (Frequentist) and marginalizing (Bayesian) the nuisance parameters give the same result; the largest correlation, $-0.17$, is with $A_b$: conditioning on $A_b = 0$ moves $A$ to $-0.0475$, as the fit with a charge-symmetric background does ($-0.0471$, Sec. VI).

### D. Bayes Factors

The Bayes factor compares two hypotheses through their evidences, $BF_{10} = P(D \mid H_1)/P(D \mid H_0)$, and requires proper priors. For the nested hypothesis $H_0$: $A = 0$ against $H_1$: $A$ free, with shared priors on the other parameters, it reduces to the Savage-Dickey ratio [11] (Appendix C.1),

$$BF_{01} = \frac{p(A = 0 \mid D)}{\pi(A = 0)}, \qquad \pi(A = 0) = \tfrac{1}{2}.$$

$A = 0$ lies 5.2 posterior standard deviations from the median, where the chain has essentially no samples, so $p(A = 0 \mid D)$ was taken from the Gaussian approximation of the marginal posterior: $BF_{01} \approx 1.1 \times 10^{-4}$ ($BF_{10} \approx 9 \times 10^3$; Fig. Va), decisive evidence against $A = 0$ on Jeffreys' scale [12]. The value scales with the prior width (Ockham factor): with $A \in (-0.3, 0.3)$, $BF_{01}$ drops by a factor 3.3 and the conclusion is unchanged.

The same test on the background asymmetry shows the Ockham factor at work: $A_b = -0.040 \pm 0.018$ is 2.3 standard deviations from zero, yet $BF_{01}(A_b = 0) \approx 3.5$ (Fig. Vb) (Appendix D.11), because the likelihood gain does not compensate for spreading the prior over $(-1, 1)$; unlike for $A$, the preference changes sign with the prior width (crossing at $W \approx 0.3$). Of the Bayesian results only the Bayes factor depends on the prior; the interval of $A$ does not (Sec. V C). A negative $A_b$ is plausible given the misidentified $B^{\pm} \to K^{\pm}\pi^+\pi^-$ and $\pi^{\pm}K^+K^-$ decays; $A_b$ is kept free in the nominal model and fixing it to zero is one of the systematic variants (Sec. VI).

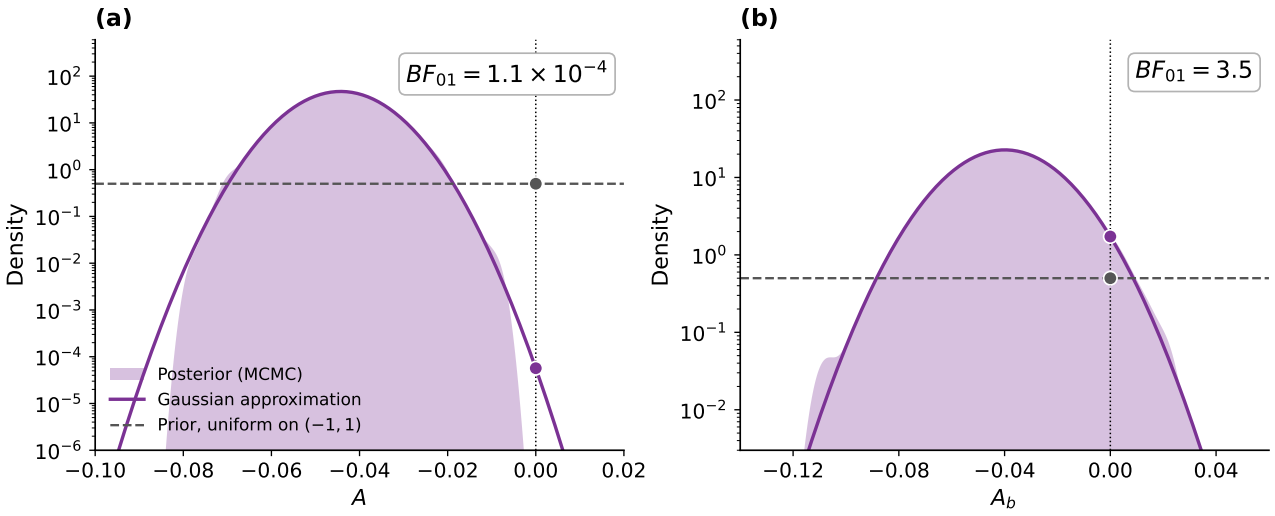



FIG. V. Savage-Dickey ratio for (a) $A$ and (b) $A_b$: marginal posterior (histogram of the chain and Gaussian approximation) and prior density, logarithmic scale. The Bayes factor $BF_{01}$ is the ratio of the two densities at zero (dots). For $A$ the posterior density at zero is $10^{-4}$ times the prior one; for $A_b$, 2.3 standard deviations from zero, it is higher than the prior density, and the data favor $A_b = 0$.

*Model Adequacy: Goodness of Fit and Posterior Predictive Check*: the model does not reach $\chi^2/\text{ndof} \approx 1$ (164.1/88), mainly because the signal tails are heavier than a Gaussian, but $A_{\text{raw}}$ changes by only $-0.0002$ with a Crystal Ball signal.

### E. Additional Results

*Sampler.* An own Metropolis-Hastings sampler reproduces the posterior of the ensemble sampler.

*Intervals.* Equal-tailed and highest-posterior-density 68% intervals coincide for $A$ and $A_b$ and differ for the skewed parameters; the profile likelihood is tabulated for combination with other results (Appendix D.7).

*Priors.* No change of the prior on the nuisance parameters or of its parametrization moves $A$ by more than 0.0002, and the evidence by direct integration agrees in order of magnitude with the Savage-Dickey ratio (Appendix D.12).

*Model adequacy.* The model does not reach $\chi^2/\text{ndof} \approx 1$ (164.1/88), because the signal tails are heavier than a Gaussian, but $A_{\text{raw}}$ changes by $-0.0002$ with a Crystal Ball signal; the posterior predictive check reproduces the asymmetry in slices of the mass ($p_B = 0.67$) and puts the difference of the $\chi^2$ of the two charges at the edge of the replicas ($p_B = 0.08$) (Appendices D.13 and D.14).

*Robustness and coverage.* $A_{\text{raw}}$ agrees between the magnet polarities ($0.7\sigma$) and the $D^0$ veto moves it by 2.3 times the expected spread; 200 toy-MC experiments give coverage $0.65 \pm 0.03$ (68%) and $0.93 \pm 0.02$ (95%) with pulls of width 1.04 (Appendices D.15 and D.16).

## VI. SYSTEMATIC UNCERTAINTIES AND RESULT

*In short: thirteen variants in four groups, fitted by both methods on the same candidates, give* 0.0070 *(Bayesian) and* 0.0072 *(Frequentist), dominated by the particle-identification requirement.* The systematic study asks how much the result depends on choices of the analysis that the data do not fix. The Frequentist and Bayesian fits share one study, so that the two uncertainties are comparable: the same thirteen variants of the nominal choices, fitted by both methods on exactly the same candidates, and the same rule to combine them. The variants (listed in Table II) change one choice at a time and fall in four groups: the *window*, the *background* model, the *signal* model and the *preselection* cuts.

The shift $\Delta = A_{\text{raw}}^{\text{var}} - A_{\text{raw}}^{\text{nom}}$ of a variant is not necessarily a systematic effect: when a variant changes the sample (window, preselection) the two samples are nested and part of $\Delta$ is a statistical fluctuation of expected size $\sigma_B = \sqrt{|\sigma^2_{\text{var}} - \sigma^2_{\text{nom}}|}$ [13], and such a variant contributes only if $|\Delta| > \sigma_B$. A variant that the data reject ($\Delta\chi^2 > 25$ on the same data) is excluded: this removes the single exponential ($\Delta\chi^2 = 134$ (Appendix D.13)). The largest remaining $|\Delta|$ of each group is its uncertainty, since the variants of a group are alternative descriptions of the same choice, and the groups, being independent choices, are added in quadrature.

Figure VI shows the scan of each particle-identification cut, with the expected spread $\sigma_B$.

The results are in Table II (Appendix E.1). The two methods give the same shifts to within $\sim$0.0005 and the same group uncertainties, with totals of 0.0070

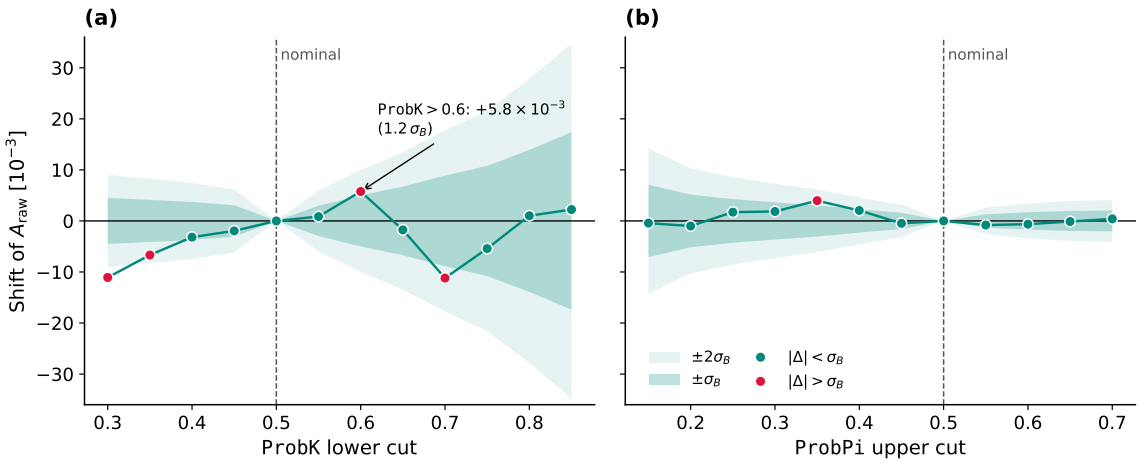



FIG. VI. Shift of $A_{\text{raw}}$ (Frequentist fit) as a function of (a) the `ProbK` lower cut and (b) the `ProbPi` upper cut, in steps of 0.05, with the expected statistical spread of the nested sample ($\pm\sigma_B$ dark, $\pm 2\sigma_B$ light; zero at the nominal cut, dash-dotted line). On the tight side of `ProbK` the points scatter around zero without trend and the pion requirement is flat; loosening `ProbK` moves $A_{\text{raw}}$ monotonically.

| Group | Variant | Bayesian | Frequentist |
|---|---|---|---|
| Window | $[5050, 5600]$ | $-0.0003$ (0.0009) | $-0.0004$ (0.0005) |
| | $[5150, 5600]$ | $-0.0001$ (0.0007) | $-0.0005$ (0.0021) |
| | $[5100, 5450]$ | $+0.0002$ (0.0009) | $-0.0001$ (0.0003) |
| | $[5100, 5700]$ | $-0.0014$ (0.0010) | $-0.0014$ (0.0001) |
| | *group uncertainty* | *0.0014* | *0.0014* |
| Background | single exponential (rejected) | $-0.0037$ | $-0.0039$ |
| | shape shared between charges | $-0.0037$ | $-0.0040$ |
| | symmetric, $A_b = 0$ | $-0.0028$ | $-0.0031$ |
| | *group uncertainty* | *0.0037* | *0.0040* |
| Signal | Crystal Ball | $-0.0002$ | $-0.0006$ |
| | $\mu, \sigma$ per charge | $-0.0000$ | $-0.0001$ |
| | *group uncertainty* | *0.0002* | *0.0006* |
| Preselection | `ProbK` $> 0.4$ | $-0.0032$ (0.0038) | $-0.0032$ (0.0037) |
| | `ProbK` $> 0.6$ | $+0.0058$ (0.0048) | $+0.0058$ (0.0050) |
| | `ProbPi` $< 0.4$ | $+0.0022$ (0.0022) | $+0.0020$ (0.0023) |
| | `ProbPi` $< 0.6$ | $-0.0004$ (0.0020) | $-0.0006$ (0.0017) |
| | *group uncertainty* | *0.0058* | *0.0058* |
| Total (quadrature) | | 0.0070 | 0.0072 |

TABLE II. Systematic study common to the Bayesian and Frequentist fits: shift $\Delta$ of $A_{\text{raw}}$ for each variant and, in parentheses and for the variants that change the sample, the expected statistical spread $\sigma_B$ of the shift. The largest contributing shift of each group is its uncertainty (italics); the groups are added in quadrature. The single exponential is rejected by the data and excluded. For the window $[5150, 5600]$ the Frequentist uncertainty is scaled from the number of events, since the Hesse matrix of that fit is not accurate.

(Bayesian) and 0.0072 (Frequentist):

$$\text{Bayesian:}$$
$$A_{\text{raw}} = -0.0443 \pm 0.0085\,(\text{stat}) \pm 0.0070\,(\text{syst}),$$
$$\text{Frequentist:}$$
$$A_{\text{raw}} = -0.0441 \pm 0.0085\,(\text{stat}) \pm 0.0072\,(\text{syst}).$$

The preselection dominates, through the single variant `ProbK` $> 0.6$ ($+0.0058$, $1.2\,\sigma_B$). The scan of each cut in steps of 0.05 (Appendix E) shows that this value is a fluctuation: on the tight side the shifts scatter around zero within $1.3\,\sigma_B$ without a trend, whereas loosening `ProbK` moves $A_{\text{raw}}$ monotonically, as expected if misidentified decays not in the mass model enter the sample. The nominal requirement is therefore well placed and the preselection uncertainty conservative. The background group is second, driven by the charge dependence of the background, and the window third.

### A. Additional Results

*The $\sigma_B$ rule.* $\sigma_B$ is the difference of two nearly equal uncertainties, so a variant at the threshold can be classified differently by the two methods without changing the result; the rule is verified by bootstrap (the spread of the shift between a sample and its tighter subsample is $\sigma_B$) (Appendix E.8).

*Stability scans.* The Frequentist grid scans shift $A_{\text{raw}}$ by at most 0.0048 over the window edges (55 of 80 fits valid), $6 \times 10^{-5}$ over the signal region and 0.0126 over the grid of the two PID cuts, whose samples are smaller (mean uncertainty 0.0112) (Appendix E.9).

## VII. VALIDATION OF THE ESTIMATOR AND OF THE INTERVALS

*In short: toy-MC experiments show an unbiased estimator at the minimum variance bound, intervals with nominal coverage and a test of $A = 0$ with the nominal size.* Toy-MC experiments generated from the fitted model of Sec. III (Poisson yields, masses drawn from the model; 1000 at the fitted value $A_{\text{true}} = -0.0441$ and 300 at each of $-0.1$, $-0.02$, 0 and $+0.05$, 1–3% of the fits not valid and excluded) verify the estimator and the profile-

likelihood ratio (Appendix F). The bias of $\hat{A}$ is at most 0.0008 (1.6 standard errors; −0.0005, −0.0001, −0.0008, +0.0002, −0.0004 for the five values), its spread equals the mean Hesse uncertainty within 6% and the pulls have width 0.94–1.05: the estimator is unbiased and reaches the minimum variance bound.

At the fitted value the coverage of $t(A_{\rm true}) \leq 1$ is 0.696±0.015 (68.3% level) and 0.954±0.007 (95.4% level); the 68% coverage over the five values is 0.71, 0.70, 0.69, 0.69 and 0.61, the last, at +0.05 (0.614±0.028), 2.5 standard deviations low, not unexpected among five values.

Inverting the Neyman construction of the same experiments at the observed $\hat{A}$ (Appendix C.2) gives the 68% interval $[-0.0528, -0.0351]$, against the Minos interval $[-0.0526, -0.0356]$; Figure VII shows bias, pull width and power as a function of the injected asymmetry.

Under $A_{\rm true} = 0$ the test $q(0) > 3.84$ rejects in $0.051 \pm 0.013$ of the experiments, the nominal size, with $q(0)$ compatible with $\chi^2_1$ (Kolmogorov-Smirnov $p = 0.79$); its power is 1.00 at the fitted value and 0.68 (0.27 at $3\sigma$) at $A_{\rm true} = -0.02$, so with 19,731 candidates an asymmetry below $\sim 0.02$ is not resolved (Appendix F.4). The $5.2\sigma$ of the data is an extrapolation of the $\chi^2_1$ tail, which the 293 experiments check only down to the $3\sigma$ level.

### A. Additional Results

*Asymptotic behavior.* With the number of candidates the bias tends to zero, the spread scales as $N^{-1/2}$ and the distribution of $\hat{A}$ becomes Gaussian (Appendix F.3).

*Goodness of fit and binning.* The binned $\chi^2$ of toy experiments follows its asymptotic distribution, so its $p$-values are accurate: the misfit of the data is 5.4 standard deviations in 10 MeV bins and 2.0 in 2.5 MeV bins, because it lives on the scale of the flanks of the peak (Appendix F.5).

## VIII. LOCAL ASYMMETRY IN THE DALITZ PLANE

*In short: the asymmetry is not uniform over the Dalitz plane ($\chi^2 = 80.8$ for 31 degrees of freedom) and is concentrated outside the $\phi(1020)$, in the region of the published analysis: $-0.234 \pm 0.025$ against $-0.239 \pm 0.020$.* To test whether the asymmetry is uniform over phase space or concentrated in specific regions, each candidate was placed in the Dalitz plane using the two opposite-sign $K^+K^-$ invariant masses $m^2(H1, H2)$ and $m^2(H1, H3)$ (Sec. II A), sorted into $m^2_{\rm lo}$ and $m^2_{\rm hi}$ to remove the $H2/H3$ labeling ambiguity. The plane $[0, 16] \times [0, 30]$ GeV$^2$ in $(m^2_{\rm lo}, m^2_{\rm hi})$ was divided into an $8 \times 8$ grid, fixed before looking at the asymmetries. In each bin with at least 20 candidates (32 of 64), the local asymmetry $A_i = (N_i^- - N_i^+)/N_i$, with binomial uncertainty $\sigma(A_i) = \sqrt{(1 - A_i^2)/N_i}$, was compared with the global counting asymmetry $A_{\rm glob} = -0.043$ through $S_i = (A_i - A_{\rm glob})/\sigma(A_i)$.

Figure VIII shows the occupancy of the bins (a), the local asymmetry (b) and its significance (c). Two of the 32 bins have $|S_i| > 3$, adjacent at $m^2_{\rm lo} < 4$ GeV$^2$ and $7.5 < m^2_{\rm hi} < 11.25$ GeV$^2$ ($S_i = -3.7$ and $-3.0$); about 0.09 such bins would be expected if the asymmetry were uniform, and two or more with probability $3 \times 10^{-3}$, a look-elsewhere number over any two of the 32 bins. The global homogeneity test, $\chi^2 = \sum_i S_i^2 = 80.8$ for 31 degrees of freedom ($p = 3 \times 10^{-6}$), confirms that the local asymmetry is not uniform; it also holds in the signal window alone, with the charm vetoes and for other grids (Appendix G).

The structure is the same with both magnet polarities: in the region of the two significant bins the counting asymmetry is $-0.129 \pm 0.023$ (magnet down) and $-0.126 \pm 0.028$ (magnet up), and the two maps agree bin by bin ($\chi^2 = 23.0$ for 29 bins, $p = 0.78$). The split tests the part of the detection asymmetry that changes sign with the polarity, not the different interaction of $K^+$ and $K^-$ with the detector material.

Where is the asymmetry in terms of the decay dynamics? The Dalitz plane shows the $\phi(1020)$ band at the kinematic edge ($m^2_{\rm lo} \approx 1.04$ GeV$^2$) and the charm bands of the $D^0$, $J/\psi$ and $\chi_{c0}$. The negative asymmetry is not produced by the $\phi(1020)$: its band (1964 candidates) has $A_{\rm count} = -0.004 \pm 0.023$, while the significant bins without it have $-0.145 \pm 0.019$. (Appendix G) It sits just above the $\phi$, where the published analysis located it, defining the region $1.2 < m^2_{\rm lo} < 2.0$ GeV$^2$, $m^2_{\rm hi} < 15$ GeV$^2$ with $A_{\rm raw} = -0.239 \pm 0.020$ [5]. The region is fixed by the paper, not chosen on our data, so the comparison is unbiased: the nominal model fitted to its 2146 candidates gives $A_{\rm raw} = -0.234 \pm 0.025$ (Bayesian) and $-0.237 \pm 0.025$ (Frequentist, $9.4\sigma$ from zero), reproducing the published value.

Three caveats limit the interpretation: the binning is coarse and was not optimized, the asymmetries are counts without sWeight background subtraction (Appendix J.5), and local variations can be partly instrumental, since the $K^\pm$ interaction asymmetry depends on momenta that are correlated with the Dalitz position. In summary, the global asymmetry is not uniform over phase space, and the region identified by the published analysis carries it.

### A. Additional Results

*Null tests.* Permuting the charges gives $\chi^2 = 31.5 \pm 8.4$ and never reaches 80.8 in 2000 permutations, and the phase-space simulation is consistent with no asymmetry (slightly optimistic $p$-values) (Appendices G.3 and G.4).

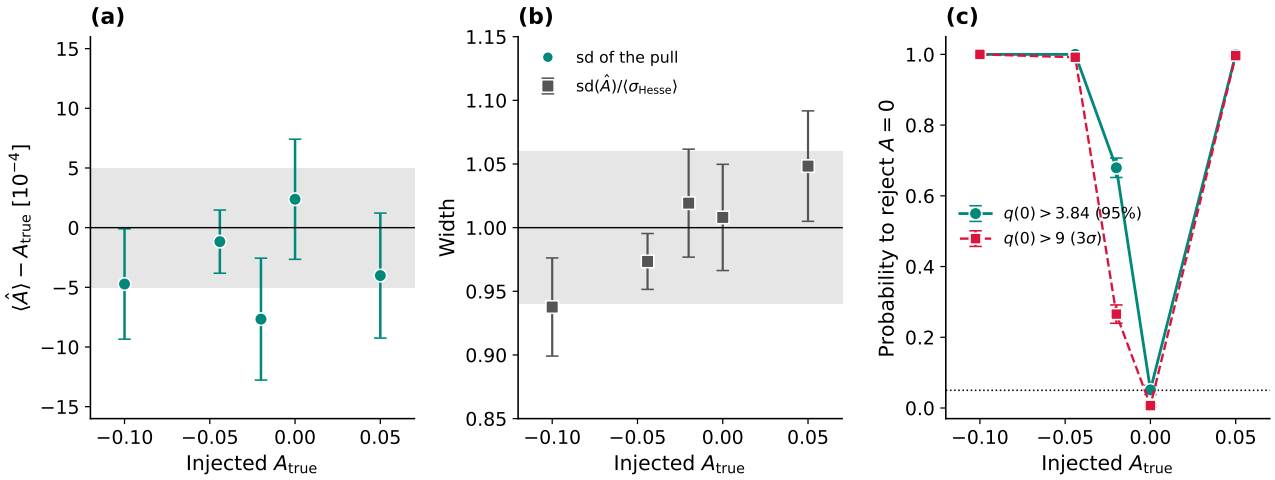



FIG. VII. Toy-MC experiments at five injected asymmetries. (a) Bias of $\hat{A}$, with its standard error. (b) Spread of the estimates divided by the mean Hesse uncertainty, and width of the pulls. (c) Probability to reject $A = 0$ in the test with $q(0) > 3.84$ (size 5%) and $q(0) > 9$ ($3\sigma$), as a function of the injected value: size at $A_{\rm true} = 0$, power elsewhere.

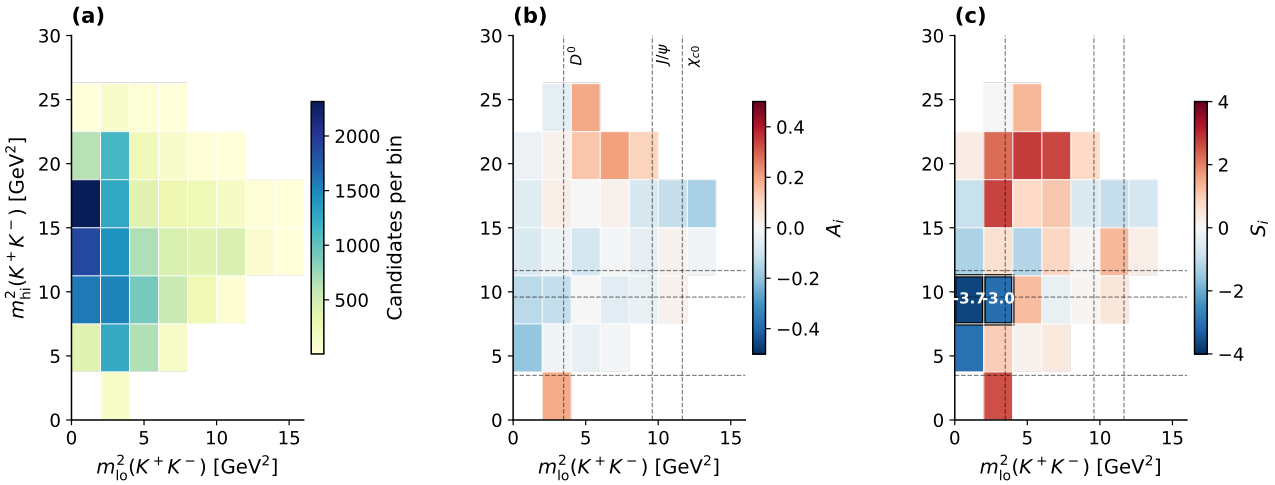



FIG. VIII. Fixed 8×8 Dalitz-plot binning, all candidates in the fit window: (a) number of candidates per bin, (b) local asymmetry $A_i$ for bins with $\geq 20$ candidates, and (c) local significance with respect to the global asymmetry, $(A_i - A_{\rm glob})/\sigma(A_i)$. The two bins with $|S_i| > 3$ are outlined. Dashed lines: the squared masses of the $D^0$, $J/\psi$ and $\chi_{c0}$, in either opposite-sign pair.

## IX. DISCUSSION AND CONCLUSIONS

*In short: the Frequentist and Bayesian analyses give the same* $A_{\rm raw} = -0.044 \pm 0.009\,(stat) \pm 0.007\,(syst)$, *5.2 standard deviations from zero, with validated intervals; they differ in the meaning of the numbers and of the tests, not in their values.* On this problem the Frequentist profile-likelihood fit and the Bayesian posterior are equivalent in value, uncertainty and coverage: with a common selection and mass model they give $A_{\rm raw} = -0.0441 \pm 0.0085 \pm 0.0072$ and $-0.0443 \pm 0.0085 \pm 0.0070$ (statistical, systematic), a nonzero asymmetry at 5.2 standard deviations, stable under the thirteen variants (the dominant systematic contribution is the particle-identification requirement) and concentrated in the Dalitz region of the published analysis outside the $\phi(1020)$ ($-0.234 \pm 0.025$ against $-0.239 \pm 0.020$).

The numbers coincide, the statements do not. The credible interval states that $A$ lies in $[-0.0528, -0.0358]$ with 68% probability given the data and the prior; the confidence interval $[-0.0526, -0.0356]$ is the set of values that the data do not reject, and its 68% is a property of the procedure that the bootstrap and the toy experiments verified (Appendix H). The Frequentist test of $A = 0$, $p = 2 \times 10^{-7}$, is independent of the prior; the Bayes factor, $BF_{10} \approx 9 \times 10^3$, depends on its width (Sec. V D), and for $A_b = -0.040 \pm 0.018$ a Frequentist test would call $2.3\sigma$ a hint while $BF_{01} \approx 3.5$ mildly favors $A_b = 0$. The Frequentist fit adds an unbiased estimator at the minimum variance bound; the Bayesian analysis adds sampling di-

agnostics and a posterior predictive check. Figure IX shows the profile likelihood and the marginal posterior of $A$ on the same axis.

The published analysis [5] finds $-0.060 \pm 0.007$ on a larger sample with a richer model (Appendix H); with the same $D^0$ veto the Bayesian fit gives $-0.0496 \pm 0.0088$, within $0.93\sigma$. The raw asymmetry is not the $CP$ asymmetry, $A_{CP} \approx A_{\text{raw}} - A_P - A_D$ (Eq. (2)): the $5.2\sigma$ here is the significance of a nonzero raw asymmetry, not of $CP$ violation. Three limitations bound the result (Appendix J.5): no tested model reaches $\chi^2/\text{ndof} \approx 1$, although $A_{\text{raw}}$ does not depend on the signal shape; no production or detection correction is applied; and the local Dalitz-plane asymmetries are counts without background subtraction. A machine-learning selection leaves $A_{\text{raw}}$ unchanged within its statistical spread (Appendix I), and the natural next step is a control-channel correction.

### A. Additional Results

*Numerical agreement.* The estimates of the two fits differ by 0.0002, both uncertainties are 0.0085, and sideband subtraction gives $-0.0503 \pm 0.0083$ (Appendix H.1).

*Profile, posterior and polarities.* Profiling and marginalizing the eleven nuisance parameters give the same distribution of $A$, since $A$ is almost uncorrelated with them; the two polarities agree and their combined likelihood reproduces the joint fit; the 68% and 95% regions in $(A, A_b)$ of the profile ($\Delta$NLL at 2.30 and 5.99) and of the highest posterior density coincide, with $A = 0$ outside both (Appendices H.3 and H.4).

*Tests and goodness of fit.* The exact binomial test on all candidates gives $6.1\sigma$. The same fit has $p = 0.03$ in 2.5 MeV bins and $p \approx 5 \times 10^{-6}$ in 10 MeV bins: the $p$-value of the binned $\chi^2$ depends on the binning as much as on the model (Appendix H).

*Low statistics.* Credible and confidence intervals coincide in value and coverage from $\sim 2 \times 10^4$ down to 50 candidates (Appendix H.11).

*Published analysis.* It has $22{,}119 \pm 164$ signal candidates, a Cruijff signal and misidentified and partially reconstructed backgrounds; its corrections move $-0.060$ to $-0.043 \pm 0.009 \pm 0.003 \pm 0.007$ and cannot be assumed for another selection (Appendix H).

*Machine learning: labels and models.* A classifier separates the signal-enriched window from the sidebands (15,130 and 1940 candidates, 20 features, 60/20/20 split) and, by the Neyman-Pearson argument of CWoLa, is also the signal/background classifier; the tree ensembles and the neural network reach test AUC 0.857–0.864, 0.09 above the linear models and indistinguishable within 0.012, with gradient boosting as reference (Appendices I.4 and I.5).

*Machine learning: regularization and uncertainty.* The linear model does not overfit, gradient boosting overfits beyond $\sim 150$ stages without losing test AUC, and class weights bias the probabilities, corrected by isotonic recalibration (Brier score $0.122 \to 0.062$); the mean per-candidate uncertainties are 0.178 (Random Forest), 0.095 (MC dropout) and 0.050 (deep ensemble) (Appendices I.6 and I.7).

*Machine learning: Dalitz plane and selection.* K-means with $K = 8$ finds two clusters beyond $3\sigma$ ($\chi^2 = 46.4$ for 7 degrees of freedom), the pattern of the fixed binning (80.8/31): negative at low $m^2_{\text{lo}}$, positive at high $m^2_{\text{hi}}$. A cut at 0.5 on the score keeps 92% of the signal and 36% of the background and gives $-0.0451 \pm 0.0087$ (Bayesian) and $-0.0447 \pm 0.0086$ (Frequentist), within the spread of a subsample, with counting too; the hyperparameter search gains at most 0.005 in AUC (winner's curse), and the selection cannot remove the misidentified background (Appendices I.8 and I.9).

## Appendix A: Dataset and Preselection

*In short: the record and its branches, the cut flow of the preselection, and the candidates and charge combinations that the main text treats in one line.*

**1. Description of the Record** *In short: the record holds* 8,556,118 *candidates of* $B^\pm \to h^\pm h^+ h^-$ *collected in 2011, in two files by magnet polarity, plus a detector-free phase-space simulation of* $B^\pm \to K^\pm K^+ K^-$ *used only to test the local-asymmetry procedure.*

**2. Branches and Preselection** *In short: the analysis reads momentum, particle-identification likelihoods, muon flag and charge of each track; the common preselection keeps* 23,895 *candidates (*0.28%*), mostly through the kaon-identification requirement.*

**3. Cut Flow of the Preselection** *In short: the kaon-identification requirement on the three tracks is by far the most selective.* The result is shown in Table III.

| Requirement | alone | cumulative | down | up |
|---|---|---|---|---|
| All candidates | – | 8,556,118 | 5,135,823 | 3,420,295 |
| ¬`isMuon`, three tracks | 6,311,517 | 6,311,517 | 3,806,297 | 2,505,220 |
| `ProbPi` $< 0.5$, three tracks | 975,724 | 481,101 | 288,908 | 192,193 |
| `ProbK` $> 0.5$, three tracks | 28,812 | 23,895 | 14,178 | 9717 |
| Mass window [5100, 5600] MeV | – | 19,731 | 11,636 | 8095 |

TABLE III. Number of candidates passing each requirement alone and after the cumulative selection, in the order of the table, and by magnet polarity. The kaon-identification requirement on the three tracks is by far the most selective.

**4. Candidates Sharing a Track** *In short: only* 42 *candidates of the fit window (*0.2%*) share a track with another candidate, a negligible effect, and all are kept.*

**5. Charge Combinations** *In short: $H1$ is always the kaon of charge opposite to the $B$ and $H2$, $H3$ the same-sign pair, so the Dalitz variables are the two opposite-sign invariant masses.*

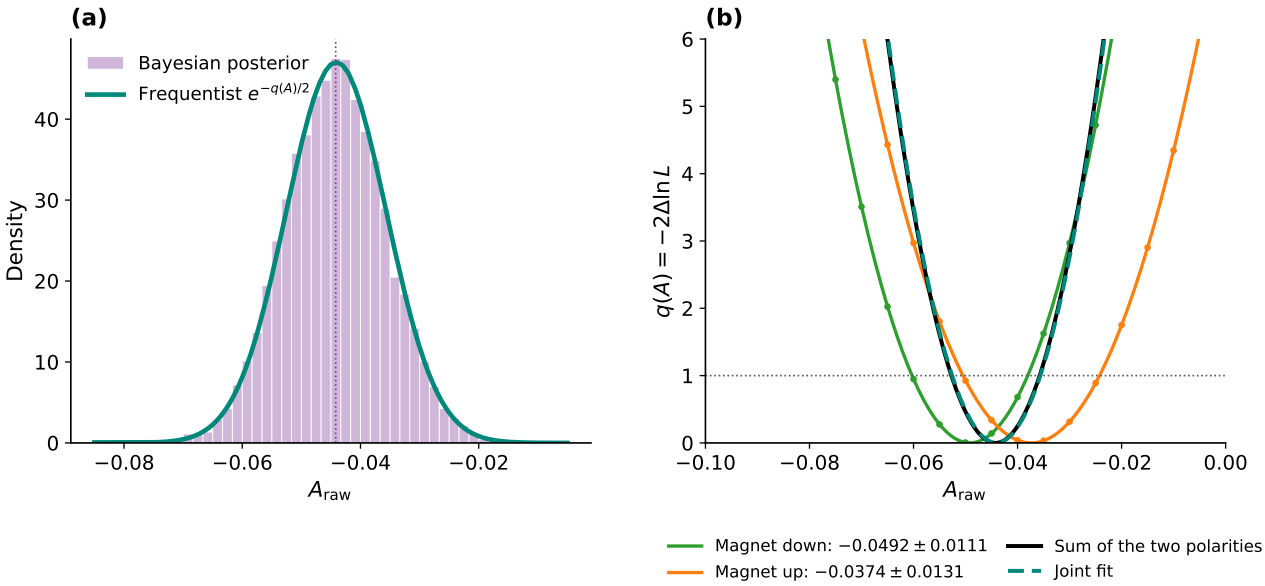



FIG. IX. (a) Frequentist profile likelihood, $\exp[-q(A)/2]$ normalized, and Bayesian marginal posterior of $A$. (b) Profile curves $q(A)$ of the two magnet polarities (parabolic fits to the points), their sum and the joint fit.

**6. Invariant Mass and Fit Window** *In short: the mass is computed with the kaon hypothesis on all tracks; the fit window* $[5100, 5600]$ *MeV holds* 19,731 *candidates and includes the start of the rise of the background below* $\sim 5150$ *MeV.*

### Appendix B: Mass Model and Likelihood

*In short: the extended likelihood and its relation to a mixture, the label switching of the two background slopes, and the six-parameter model that preceded the nominal one.*

**1. Extended Likelihood** *In short: with a Poisson total number of candidates the likelihood is* $e^{-(N_s+N_b)} \prod_i [N_s S + N_b B]$ *up to a constant, which constrains the yields and* $A$ *together.*

**2. Extended Likelihood versus a Mixture with Fixed Total** *In short: the extended form differs from a mixture with fixed total only by the Poisson term of the total yield, which carries no information on* $A$*; the purity of the signal window is about* 96%*.*

**3. Label Switching of the Background Components** *In short: the mixture is invariant under the exchange of the two exponentials, which would give two identical modes; the ordering* $\lambda_1 > \lambda_2$ *removes the symmetry.*

**4. Minimal Model as a Cross-Check** *In short: a six-parameter model on a narrower window gives* $A_{\text{raw}} = -0.0497$ *(68% CI* $[-0.058, -0.041]$*), compatible with the nominal result within its uncertainty; it is the model used for the low-statistics comparison.*

### Appendix C: Frequentist Analysis

*In short: what the Hesse and Minos uncertainties and the test of* $A = 0$ *are, the Neyman construction, the uncertainties of the twelve parameters by three methods, and the behavior of the fit from different starting values.*

**1. Hesse Uncertainties, Test of $A = 0$ and Significance** *In short: the Hesse matrix is the inverse of the Fisher information, the test statistic is the profile likelihood ratio, and the significance is* $Z = \sqrt{q(0)}$*.*

**2. Neyman Construction and Dependence on the Momentum** *In short: inverting the test of* $A_{\text{true}}$ *with toy-MC experiments gives the same interval as the likelihood ratio, and* $A_{\text{raw}}$ *does not depend on the momentum of the* $B$*.* The result is shown in Figure X.

The result is shown in Table IV.

| Bin | range [GeV] | $n$ | $A_{\text{raw}}$ |
|---|---|---|---|
| $p_T$, 1 | 1.0–4.7 | 6577 | $-0.0563 \pm 0.0169$ |
| $p_T$, 2 | 4.7–8.2 | 6577 | $-0.0362 \pm 0.0141$ |
| $p_T$, 3 | 8.2–30.5 | 6577 | $-0.0452 \pm 0.0134$ |
| $p$, 1 | 26.7–78.1 | 6577 | $-0.0391 \pm 0.0139$ |
| $p$, 2 | 78.1–110.1 | 6577 | $-0.0475 \pm 0.0149$ |
| $p$, 3 | 110.1–288.5 | 6577 | $-0.0478 \pm 0.0153$ |

TABLE IV. $A_{\text{raw}}$ (joint frequentist fit) in terciles of the transverse momentum and of the momentum of the $B$ candidate.

The result is shown in Figure XI.

**3. Uncertainties of the Twelve Parameters by Three Methods** *In short: the inverse of the Fisher information (Hesse), the bootstrap of the data and the*

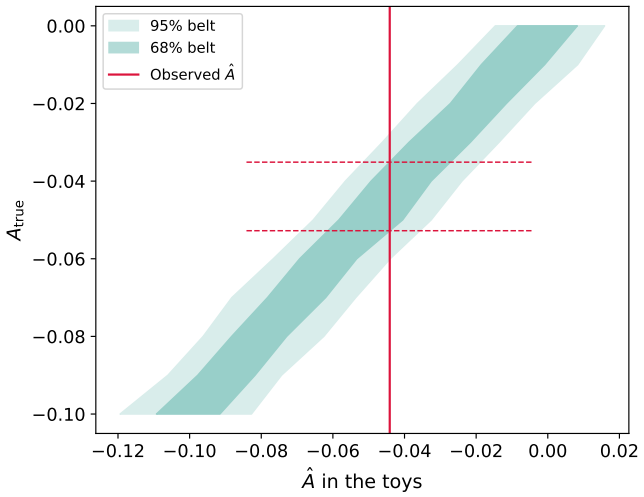



FIG. X. Confidence belt of $\hat{A}$ from toy-MC experiments (central 68% and 95% ranges) as a function of $A_{\text{true}}$; the vertical line is the observed $\hat{A}$ and the dashed lines the resulting 68% interval.

*posterior give the same standard deviations for $A$, $A_b$, $\mu$ and $\sigma$; the bootstrap differs for the two yields, because it keeps the total number of candidates fixed, and for the steep slope $\lambda_1^+$, which is weakly constrained.* The result is shown in Table V.

| Parameter | Hesse | Bootstrap | Posterior |
|---|---|---|---|
| $A$ | 0.0085 | 0.0085 | 0.0085 |
| $A_b$ | 0.0177 | 0.0183 | 0.0176 |
| $N_s$ | 133 | 72 | 133 |
| $N_b$ | 82 | 72 | 82 |
| $\mu$ [MeV] | 0.159 | 0.159 | 0.161 |
| $\sigma$ [MeV] | 0.141 | 0.150 | 0.138 |
| $\lambda_1^+$ | 0.0077 | 0.0057 | 0.0078 |
| $\lambda_2^+$ | 0.00028 | 0.00030 | 0.00029 |
| $f_+$ | 0.028 | 0.029 | 0.030 |
| $\lambda_1^-$ | 0.0047 | 0.0043 | 0.0047 |
| $\lambda_2^-$ | 0.00040 | 0.00038 | 0.00039 |
| $f_-$ | 0.039 | 0.036 | 0.037 |

TABLE V. Standard deviation of the parameters of the nominal fit from the Hesse matrix, from the bootstrap of the data and from the posterior.

The result is shown in Figure XII.

**4. Starting Values and Weakly Constrained Parameters** *In short: 29 of 30 fits from random starting points reach the same minimum; the steep-background fraction is the weakly constrained direction, as the long chain of the Bayesian fit already showed.*

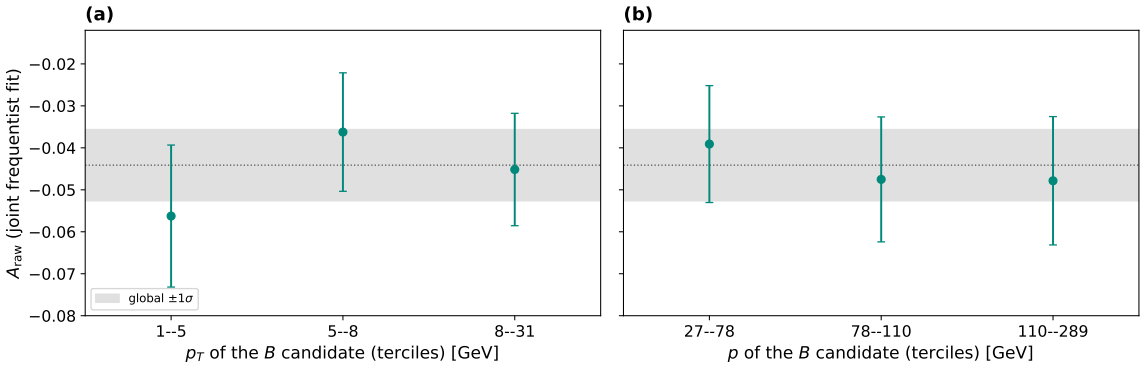



FIG. XI. $A_{\text{raw}}$ (joint frequentist fit) in terciles of (a) the transverse momentum and (b) the momentum of the $B$ candidate, with the global value and its $\pm 1\sigma$ band; each bin has 6577 candidates.

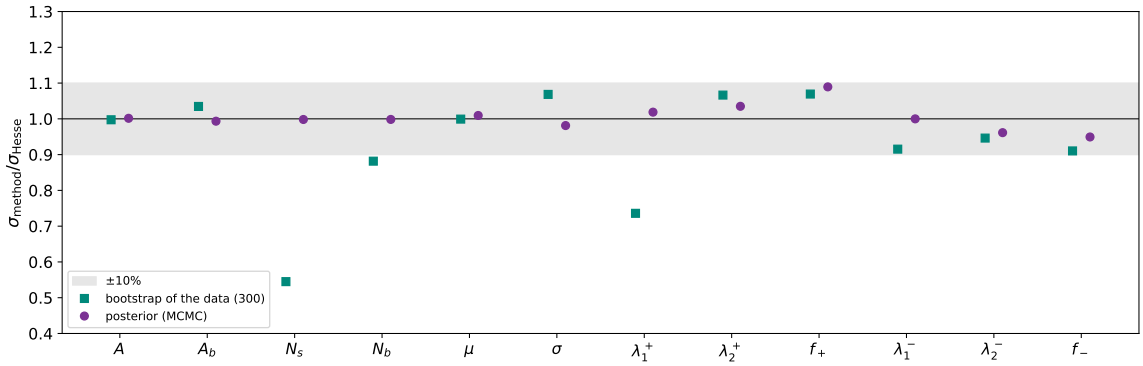



FIG. XII. Standard deviation of the twelve parameters of the nominal fit from the bootstrap of the data (300 replicas) and from the posterior, divided by the Hesse uncertainty; the band is ±10%.

## Appendix D: Bayesian Analysis

*In short: the sampling and the posterior summaries behind Sec. V, the Bayes factors with their derivation, the model-adequacy and robustness checks, the toy-MC coverage of the credible interval and the alternative sampler.*

**1. Sampling the Posterior** *In short: Metropolis-Hastings accepts a proposal with a ratio that satisfies detailed balance; the affine-invariant ensemble move samples correlated posteriors without tuning, and $N/\tau$ is the effective length of a chain.*

**2. Posterior Summaries and Model Comparison** *In short: equal-tailed and HPD intervals, the Jeffreys scale for Bayes factors, the Ockham effect of the prior width and the definition of the posterior predictive p-value.*

**3. Flat Priors and the Evidence** *In short: a flat prior is not invariant under reparametrization, but with $\sim 2\times10^4$ events the posterior is dominated by the likelihood; the evidence is a constant that neither the MCMC nor the Savage-Dickey ratio needs.*

**4. Sampling and Convergence Diagnostics** *In short: 32 walkers for 25,000 steps give $N/\tau \geq 113$ autocorrelation times for every parameter, and runs started over the whole prior box converge to the same single mode.*

**5. Trace Plots of the Walkers** *In short: walkers started over the whole prior box converge to the same mode.* The result is shown in Figure XIII.

**6. Posterior of the Raw Asymmetry** *In short: $A_{\text{raw}} = -0.0443$ with 68% and 95% intervals $[-0.0528, -0.0358]$ and $[-0.0608, -0.0276]$; the posterior is Gaussian, so equal-tailed and highest-density intervals coincide and its mode is the maximum-likelihood estimate.*

**7. Equal-Tailed and Highest-Posterior-Density Intervals** *In short: for $A$ and $A_b$ the two kinds of 68% interval coincide; they differ for the skewed parameters, whose highest-posterior-density interval is shifted towards the mode.* The result is shown in Table VI.

| Parameter | median | equal-tailed | HPD | skewness |
|---|---|---|---|---|
| $A$ | $-0.0442$ | $[-0.0529, -0.0358]$ | $[-0.0525, -0.0355]$ | $+0.01$ |
| $A_b$ | $-0.0400$ | $[-0.0575, -0.0224]$ | $[-0.0575, -0.0225]$ | $+0.01$ |
| $f_+$ | 0.114 | $[0.089, 0.145]$ | $[0.084, 0.138]$ | $+0.90$ |
| $f_-$ | 0.209 | $[0.174, 0.248]$ | $[0.168, 0.241]$ | $+0.33$ |
| $\lambda_1^+$ [MeV$^{-1}$] | 0.0350 | $[0.0281, 0.0431]$ | $[0.0274, 0.0423]$ | $+0.52$ |

TABLE VI. Equal-tailed and highest-posterior-density 68% intervals and skewness of the marginal posteriors.

The result is shown in Figure XIV.

**8. Savage–Dickey Ratio** *In short: the Bayes factor of $A = 0$ is the ratio of the posterior and prior densities at $A = 0$, with no integral over the twelve parameters.*

**9. Bayes Factor of $A = 0$** *In short: the Savage-Dickey ratio with the Gaussian approximation of the posterior gives $BF_{01} \approx 1.1\times10^{-4}$ ($BF_{10} \approx 9\times10^3$), decisive evidence against $A = 0$ that scales with the prior width.*

**10. Bayes Factor of the Background Asymmetry** *In short: $A_b = -0.040 \pm 0.018$ is 2.3 standard deviations from zero, yet $BF_{01} \approx 3.5$ favors $A_b = 0$ (Ockham factor); $A_b$ is kept free in the nominal model.*

**11. Savage–Dickey Densities of the Parameters** *In short: the posterior and prior densities at the null value, which give the Bayes factors of the main text.*

**12. Priors and Bayes Factors** *In short: no change of the prior on the nuisance parameters or of its parametrization moves $A$ by more than 0.0002; the Bayes factor of $A = 0$ scales with the prior width, and the evidence computed by direct integration agrees in order of magnitude with the Savage-Dickey ratio.* The result is shown in Table VII.

| Prior | $A$ (median) | 68% interval | ESS |
|---|---|---|---|
| Nominal (uniform) | $-0.0442$ | $[-0.0529, -0.0358]$ | 20000 |
| Log-uniform $N_s$, $N_b$ | $-0.0442$ | $[-0.0529, -0.0358]$ | 19993 |
| Log-uniform slopes | $-0.0440$ | $[-0.0528, -0.0356]$ | 12306 |
| Uniform in $1/\lambda$ | $-0.0444$ | $[-0.0526, -0.0360]$ | 153 |
| Flat in $(N_s^+, N_s^-)$ | $-0.0442$ | $[-0.0529, -0.0358]$ | 19999 |
| Flat in $(N_s^\pm)$ and $(N_b^\pm)$ | $-0.0442$ | $[-0.0529, -0.0358]$ | 19993 |

TABLE VII. Posterior of $A$ under alternative priors, by importance reweighting of the chain; ESS: effective sample size of the 20,000 thinned samples.

**13. Model Adequacy: Goodness of Fit and Posterior Predictive Check** *In short: the model does not reach $\chi^2/\text{ndof} \approx 1$ (164.1/88), mainly because the signal tails are heavier than a Gaussian, but $A_{\text{raw}}$ changes by only $-0.0002$ with a Crystal Ball signal.* The result is shown in Table VIII.

The result is shown in Figure XV.

The result is shown in Figure XVI.

**14. Posterior Predictive Check with Statistics Not Fitted by the Model** *In short: the asymmetry in slices of the mass, which the model predicts, is reproduced ($p_B = 0.67$), and the difference between the $\chi^2$ of the two charges is at the edge of the replicas ($p_B = 0.08$).* The result is shown in Figure XVII.

**15. Robustness Checks: Polarity, Prior and Charm Vetoes** *In short: $A_{\text{raw}}$ agrees between the two magnet polarities ($0.7\sigma$) and does not depend on the prior; the $D^0$ veto moves it by 2.3 times the expected statistical spread.* The result is shown in Figure XVIII.

**16. Toy-MC Coverage Validation** *In short: 200 toy-MC experiments give coverage $0.65 \pm 0.03$ (68%) and $0.93 \pm 0.02$ (95%), consistent with nominal, with unbiased estimates and pulls of width 1.04.* The result is shown in Figure XIX.

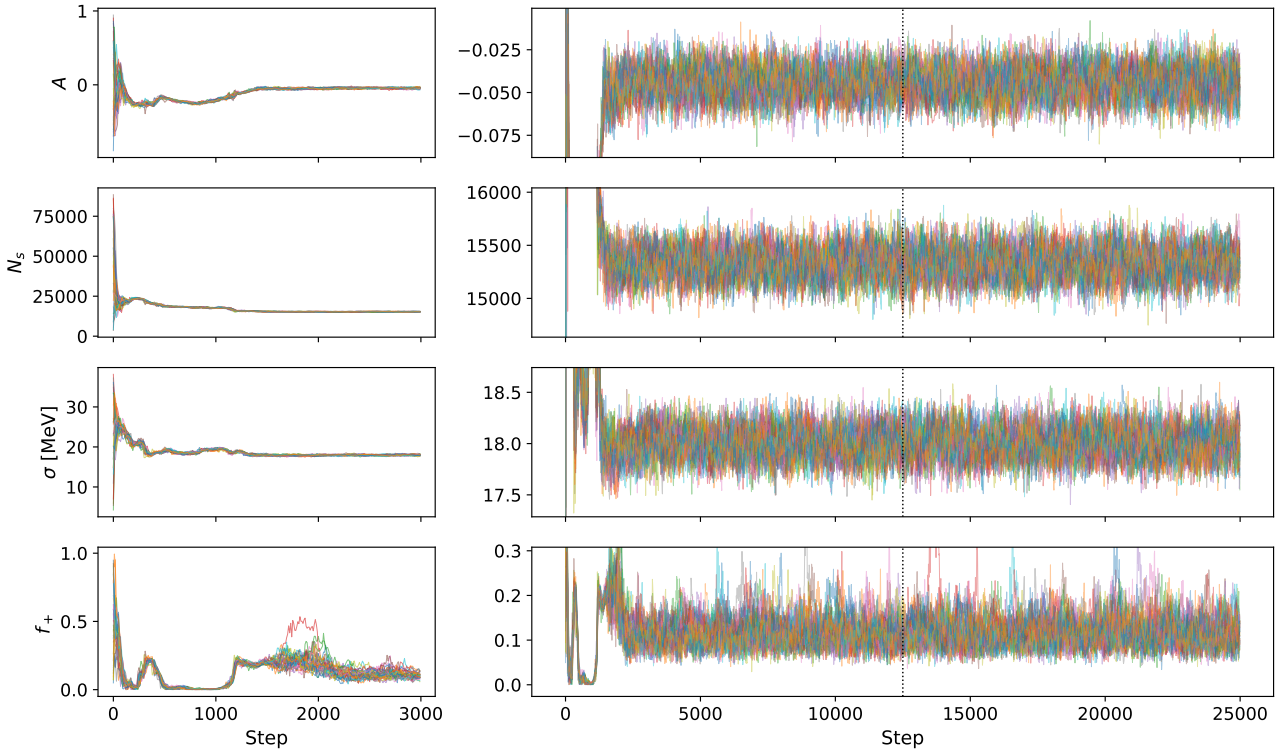



FIG. XIII. Trace plot of the nominal fit with the 32 walkers initialized uniformly over the prior box (one line per walker, every 10th step), for $A$, $N_s$, $\sigma$ and the steep-background fraction $f_+$. Left: first 3000 steps, full range; right: whole chain, zoomed on the posterior. The first half of the chain (dotted line) is discarded as burn-in.

| Model | $A_{\mathrm{raw}}$ | $\chi^2$/ndof | $p$ |
|---|---|---|---|
| Nominal | $-0.0443$ | 164.1/88 | $2 \times 10^{-6}$ |
| Crystal Ball signal | $-0.0445$ | 131.0/86 | $1 \times 10^{-3}$ |
| $\mu$, $\sigma$ per charge | $-0.0443$ | 164.2/86 | $8 \times 10^{-7}$ |
| Shared bkg. shape | $-0.0480$ | 174.1/91 | $4 \times 10^{-7}$ |
| Symmetric bkg. ($A_b = 0$) | $-0.0471$ | 169.8/89 | $5 \times 10^{-7}$ |
| Single exponential | $-0.0480$ | 297.7/92 | $2 \times 10^{-23}$ |

TABLE VIII. $A_{\mathrm{raw}}$ (posterior median) and goodness of fit of the nominal model (Gaussian signal, double exponential per charge) and of the alternative signal and background models, same data and window.

## 17. Metropolis-Hastings Sampler and Published Likelihood

*In short: an own Metropolis-Hastings sampler reproduces the posterior of the ensemble sampler, and the profile likelihood is given in a table, so that the result can be combined with others.* The result is shown in Table IX.

| $A$ | $-0.08$ | $-0.07$ | $-0.06$ | $-0.05$ | $-0.04$ | $-0.03$ | $-0.02$ | $-0.01$ | $+0.00$ |
|---|---|---|---|---|---|---|---|---|---|
| $q(A)$ | 17.9 | 9.3 | 3.5 | 0.5 | 0.2 | 2.8 | 8.1 | 16.1 | 27.0 |

TABLE IX. Profile likelihood ratio $q(A) = -2\Delta \ln L$ of the nominal fit (other parameters profiled); minimum at $-0.0441$, Minos intervals $[-0.0526, -0.0356]$ (68%) and $[-0.0607, -0.0275]$ (95%).

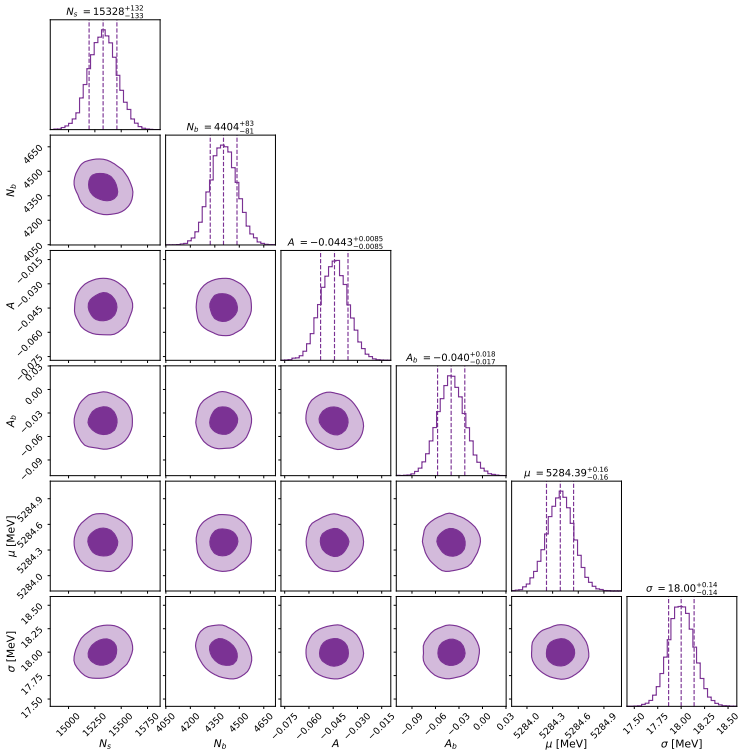



FIG. XIV. Posterior of the six physics parameters of the nominal fit (the six background-shape parameters are marginalized). Diagonal: marginal posteriors with the 16th, 50th and 84th percentiles (dashed lines and titles). Off-diagonal: 2D marginals, contours enclosing the regions of the $1\sigma$ and $2\sigma$ levels of a 2D Gaussian (39% and 86%).

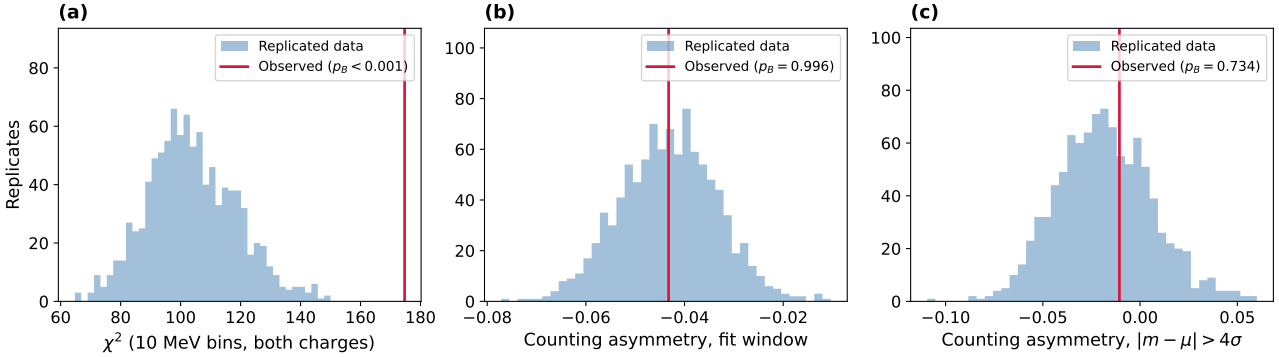



FIG. XV. Posterior predictive check: distributions of three discrepancies over 1000 datasets replicated from posterior draws, compared with the observed values. (a) binned $\chi^2$; (b) and (c) counting asymmetry of the fit window and of the sidebands.

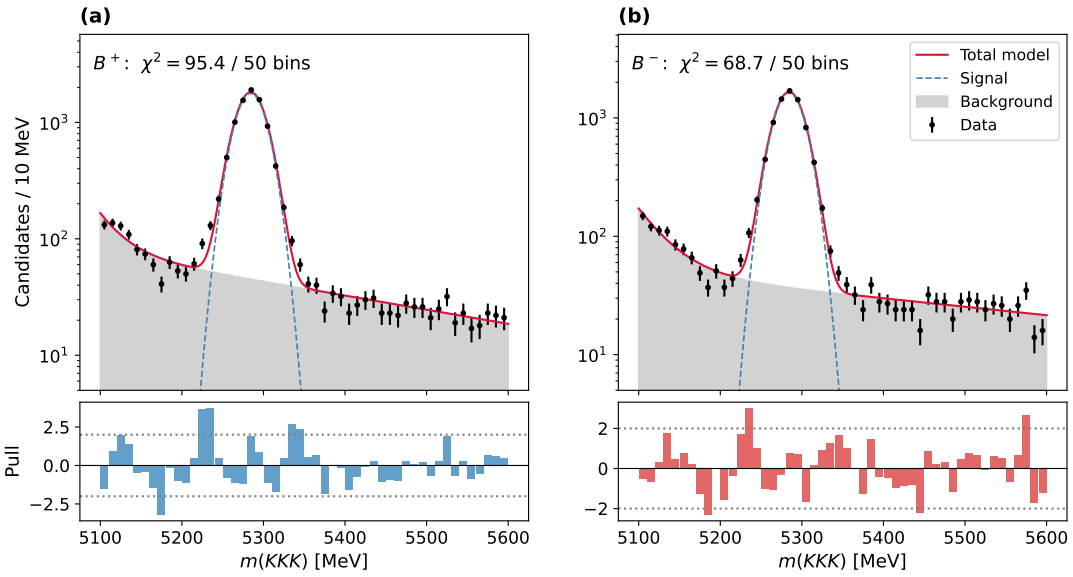



FIG. XVI. Mass distributions of (a) $B^+$ and (b) $B^-$ candidates with the nominal model (posterior medians, logarithmic scale): total, signal and background components. Bottom: pulls, $(\text{obs} - \text{exp})/\sqrt{\text{exp}}$.

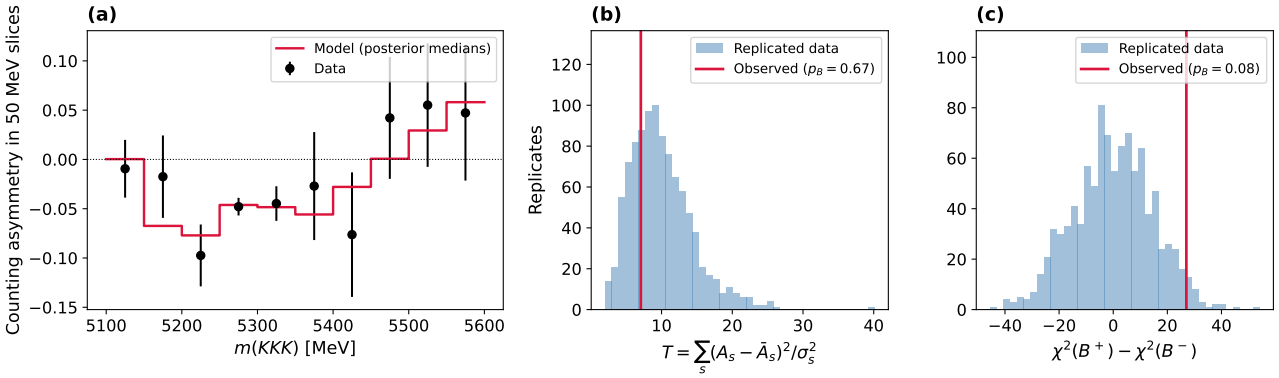



FIG. XVII. (a) Counting asymmetry in slices of 50 MeV of the mass, data and model. (b) Distribution of $T = \sum_s (A_s - \bar{A}_s)^2/\sigma_s^2$ over 1000 replicas and observed value. (c) The same for $\chi^2(B^+) - \chi^2(B^-)$.

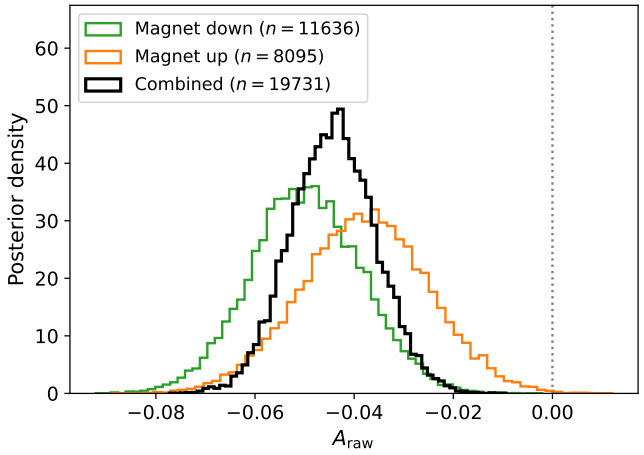



FIG. XVIII. Posterior of $A_{\mathrm{raw}}$ for the magnet-down, magnet-up and combined samples.

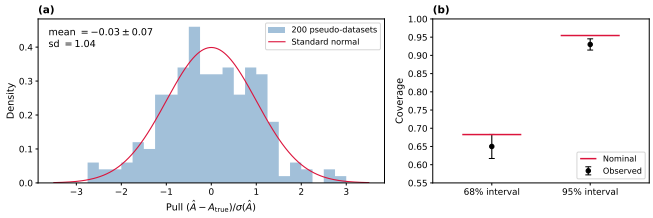



FIG. XIX. Toy-MC validation, 200 toy-MC datasets. (a) Pulls $(\hat{A} - A_{\mathrm{true}})/\sigma(\hat{A})$, with $\hat{A}$ and $\sigma(\hat{A})$ the posterior median and standard deviation of each replica, compared with a standard normal. (b) Fraction of replicas whose 68% and 95% credible intervals contain the injected value, with binomial uncertainties, and the nominal coverage.

### Appendix E: Systematic Uncertainties

*In short: every fit of the nominal model and of its variants with both methods, the variants and the rule that combines them, and the validation of that rule on nested samples.*

**1. All Fits of the Nominal Model and of Its Variants** *In short: the Bayesian and frequentist results of every variant and cross-check side by side.* The result is shown in Table X.

**2. List of the Variants** *In short: thirteen variants of the nominal choices, in four groups (window, background, signal, preselection), fitted by both methods on exactly the same candidates.*

**3. Combination Rule** *In short: a variant that changes the sample counts only if its shift exceeds $\sigma_B$; variants rejected by the data are excluded, the largest shift of each group is its uncertainty, and the groups are added in quadrature.*

**4. Results and Particle-Identification Scan** *In short: the total systematic uncertainties are* 0.0070 *(Bayesian) and* 0.0072 *(frequentist), dominated by the preselection through the single variant* `ProbK` $>$ 0.6 *(+0.0058, 1.2 $\sigma_B$).*

**5. Shifts of All Variants** *In short: the shifts of the Bayesian and frequentist fits coincide variant by variant; the preselection group dominates, and the single exponential, rejected by the data, is drawn open.* The result is shown in Figure XX.

**6. Scan of the Particle-Identification Cuts** *In short: on the tight side the shift scatters around zero within 1.3 $\sigma_B$ and the pion requirement is flat; loosening* `ProbK` *moves $A_{\mathrm{raw}}$ monotonically, which supports the nominal requirement.*

**7. Sensitivity of $\sigma_B$ to Differences between the Methods** *In short: $\sigma_B$ is the difference of two nearly equal uncertainties and is sensitive to small differences between the methods; a variant at the threshold can be classified differently without changing the result.*

**8. Validation of the Rule for Nested Samples** *In short: the spread of the shift of $\hat{A}$ between a sample and its tighter-identification subsample is the one given by the quadrature difference of the uncertainties.* The result is shown in Figure XXI.

**9. Stability Scans of the Frequentist Fit: Numbers** *In short: the largest shifts are* 0.0048 *over the grid of window edges and* 0.0126 *over the grid of the two PID cuts, whose samples are smaller (mean uncertainty* 0.0112*); the signal region moves $A_{\mathrm{raw}}$ by $6 \times 10^{-5}$.*

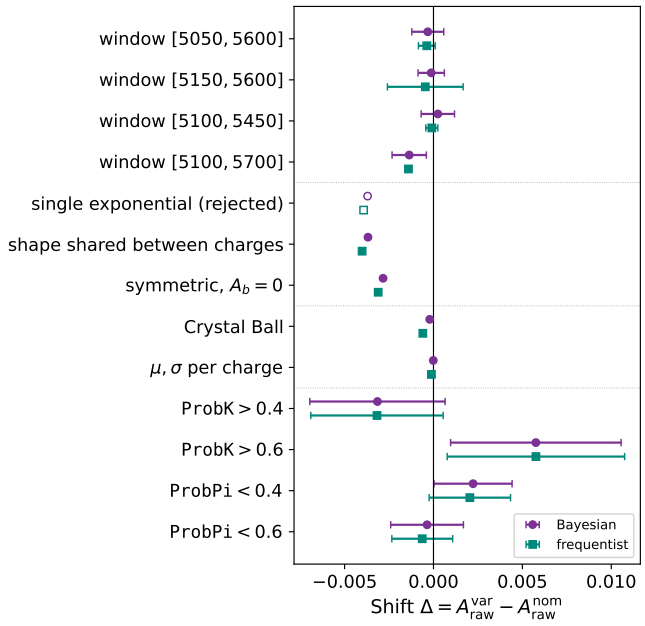



FIG. XX. Shift $\Delta = A_{\mathrm{raw}}^{\mathrm{var}} - A_{\mathrm{raw}}^{\mathrm{nom}}$ of every variant of the systematic study, Bayesian and frequentist fits; bars: expected spread $\sigma_B$ of a nested subsample (variants that change the sample only). Open symbols: variant rejected by the data. The dotted lines separate the groups (window, background, signal, preselection).

| Fit | $n$ | $A_{\text{raw}}$ Bayesian (median) | 68% CI | $\chi^2$/ndof | $A_{\text{raw}}$ frequentist |
|---|---|---|---|---|---|
| Nominal | 19731 | $-0.0443$ | $[-0.0528, -0.0358]$ | 164.1/88 | $-0.0441 \pm 0.0085$ |
| *Systematic variants (Sec. VI)* | | | | | |
| Window $[5050, 5600]$ | 21164 | $-0.0446$ | $[-0.0531, -0.0362]$ | 217.4/98 | $-0.0445 \pm 0.0085$ |
| Window $[5150, 5600]$ | 18566 | $-0.0444$ | $[-0.0528, -0.0359]$ | 141.5/78 | $-0.0446 \pm 0.0088^{\dagger}$ |
| Window $[5100, 5450]$ | 19004 | $-0.0440$ | $[-0.0524, -0.0357]$ | 136.9/58 | $-0.0442 \pm 0.0085$ |
| Window $[5100, 5700]$ | 20191 | $-0.0456$ | $[-0.0540, -0.0373]$ | 200.2/108 | $-0.0455 \pm 0.0085$ |
| Single exponential | 19731 | $-0.0480$ | $[-0.0564, -0.0395]$ | 297.7/92 | $-0.0481 \pm 0.0085$ |
| Background shape shared | 19731 | $-0.0480$ | $[-0.0564, -0.0397]$ | 174.1/91 | $-0.0481 \pm 0.0083$ |
| Symmetric background | 19731 | $-0.0471$ | $[-0.0552, -0.0388]$ | 169.8/89 | $-0.0472 \pm 0.0084$ |
| Crystal Ball signal | 19731 | $-0.0445$ | $[-0.0529, -0.0363]$ | 131.0/86 | $-0.0447 \pm 0.0085$ |
| $\mu, \sigma$ per charge | 19731 | $-0.0443$ | $[-0.0527, -0.0356]$ | 164.2/86 | $-0.0442 \pm 0.0087$ |
| `ProbK` $> 0.4$ | 26553 | $-0.0474$ | $[-0.0550, -0.0399]$ | 175.6/88 | $-0.0473 \pm 0.0076$ |
| `ProbK` $> 0.6$ | 13981 | $-0.0385$ | $[-0.0482, -0.0289]$ | 145.0/88 | $-0.0384 \pm 0.0099$ |
| `ProbPi` $< 0.4$ | 17984 | $-0.0421$ | $[-0.0506, -0.0333]$ | 150.1/88 | $-0.0421 \pm 0.0088$ |
| `ProbPi` $< 0.6$ | 20953 | $-0.0446$ | $[-0.0529, -0.0366]$ | 159.0/88 | $-0.0448 \pm 0.0083$ |
| *Cross-checks (Secs. V B, D.15)* | | | | | |
| Dispersed start, run 1 | 19731 | $-0.0446$ | $[-0.0529, -0.0360]$ | 164.1/88 | – |
| Dispersed start, run 2 | 19731 | $-0.0442$ | $[-0.0528, -0.0356]$ | 164.1/88 | – |
| Magnet down | 11636 | $-0.0497$ | $[-0.0607, -0.0386]$ | 153.1/88 | – |
| Magnet up | 8095 | $-0.0378$ | $[-0.0507, -0.0251]$ | 150.3/88 | – |
| Narrow prior, $A \in (-0.3, 0.3)$ | 19731 | $-0.0443$ | $[-0.0528, -0.0358]$ | 164.1/88 | – |
| $D^0$ veto | 18744 | $-0.0496$ | $[-0.0584, -0.0409]$ | 159.1/88 | $-0.0493 \pm 0.0087$ |
| $D^0$, $J/\psi$, $\chi_{c0}$ vetoes | 16367 | $-0.0423$ | $[-0.0516, -0.0329]$ | 140.1/88 | $-0.0421 \pm 0.0094$ |

TABLE X. All fits of the nominal model and of its variants: number of candidates, Bayesian posterior median and 68% credible interval of $A_{\text{raw}}$, binned goodness of fit (10 MeV bins), and frequentist estimate with its Hesse uncertainty (Sec. IV; fitted on the same candidates; the cross-checks of the Bayesian fits that have no frequentist counterpart in this work are marked –). $^{\dagger}$ Uncertainty scaled from the number of events, since the Hesse matrix of that fit is not accurate. Every fit uses 32 walkers and 25,000 steps.

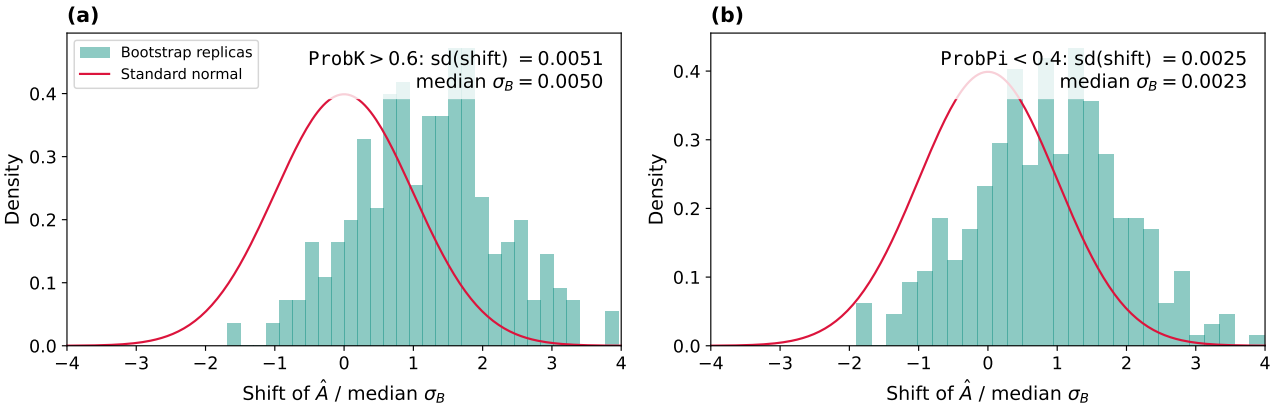



FIG. XXI. Shift of $\hat{A}$ between the nominal sample and a tighter selection, divided by the median $\sigma_B$, in 300 bootstrap replicas of the data: (a) `ProbK` $> 0.6$, (b) `ProbPi` $< 0.4$, and standard normal distribution.

The result is shown in Figure XXII.

## Appendix F: Validation of the Estimator and of the Intervals

*In short: how the toy-MC experiments are generated, the bias, efficiency and coverage of the estimator, its asymptotic behavior, the size and power of the test of $A = 0$ and the goodness of fit as a function of the binning.*

**1. Generation of the Toy-MC Experiments** *In short: toy experiments are generated from the fitted model (Poisson yields, masses by the transformation method) and fitted with the nominal procedure:* 1000 *at the fitted value and* 300 *at each of four other values of $A$.*

**2. Bias, Efficiency and Coverage** *In short: the bias is at most* 0.0008*, the spread equals the Hesse uncertainty within* 6%*, and the coverage at the fitted value is* $0.696 \pm 0.015$ *and* $0.954 \pm 0.007$*.*

**3. Asymptotic Behavior of the Estimator** *In short: with the number of candidates the bias tends to zero, the spread scales as $N^{-1/2}$ and the distribution of $\hat{A}$ becomes Gaussian, as the central limit theorem and the asymptotic properties of the maximum-likelihood estimate predict.* The result is shown in Table XI.

| $N$ | bias | sd($\hat{A}$) | sd$\cdot\sqrt{N}$ | skewness | ex. kurtosis | Shapiro $p$ |
|---|---|---|---|---|---|---|
| 50 | −0.0085 | 0.1611 | 1.14 | −0.15 | +0.64 | 0.03 |
| 150 | −0.0051 | 0.0920 | 1.13 | +0.16 | +0.31 | 0.59 |
| 500 | −0.0022 | 0.0501 | 1.12 | −0.08 | +0.38 | 0.52 |
| 2000 | +0.0009 | 0.0225 | 1.01 | −0.20 | −0.20 | 0.32 |

TABLE XI. Distribution of the maximum-likelihood estimate of $A$ in 300 toy-MC datasets of $N$ candidates (six-parameter model of Sec. B.4).

**4. Size and Power of the Test of $A = 0$** *In short: the test has the nominal size, the statistic $q(0)$ follows a $\chi^2$ distribution with one degree of freedom, and an asymmetry of $-0.02$ would be rejected at $3\sigma$ in one toy-MC experiment out of four.* The result is shown in Figure XXIII.

**5. Goodness of Fit and the Binning** *In short: the binned $\chi^2$ of toy-MC experiments follows its asymptotic distribution, so its p-values are accurate; the misfit of the data is* 5.4 *standard deviations at* 10 *MeV bins and* 2.0 *at* 2.5 *MeV, because it lives on the scale of the flanks of the peak.* The result is shown in Figure XXIV.

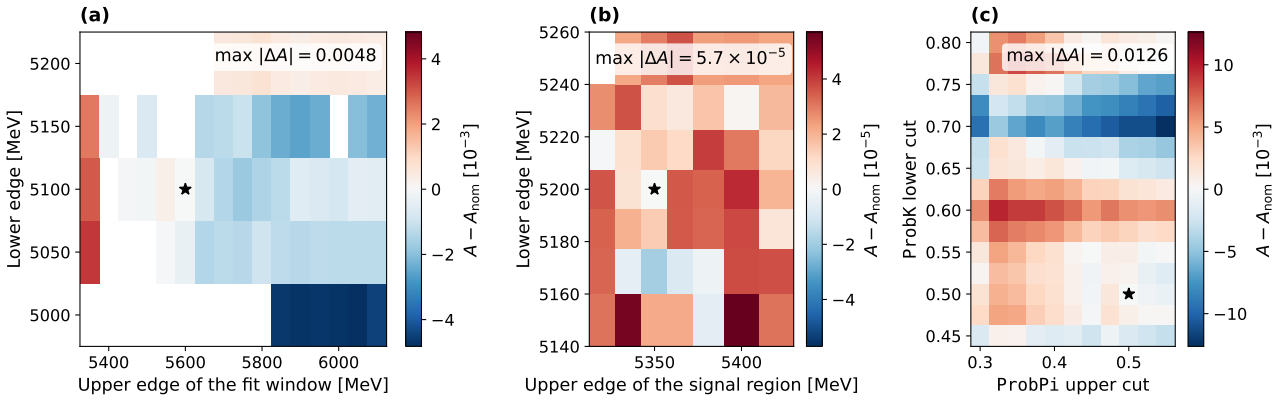



FIG. XXII. Stability of the fitted $A_{\text{raw}}$ over the three grids: (a) lower and upper edge of the fit window (only fits with a valid minimum and an accurate covariance are shown); (b) lower and upper edge of the signal region; (c) lower cut on `ProbK` and upper cut on `ProbPi`. The colour is the shift with respect to the nominal fit (star), in units of $10^{-3}$ for (a) and (c) and $10^{-5}$ for (b); the statistical uncertainty of the nominal fit is $8.5 \times 10^{-3}$.

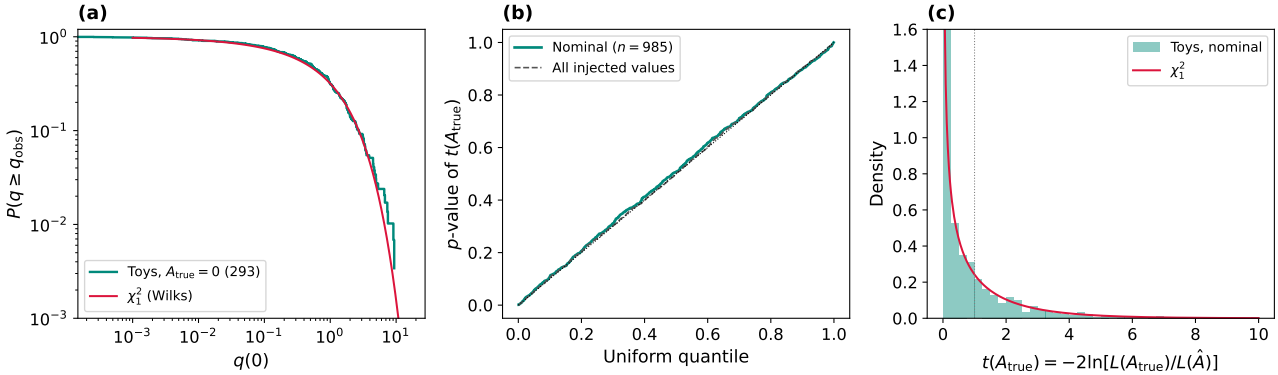



FIG. XXIII. Toy-MC experiments, validity of Wilks' theorem. (a) Fraction of experiments with $q(0)$ above the observed value, for $A_{\text{true}} = 0$, compared with the $\chi^2_1$ distribution. (b) $p$-value of $t(A_{\text{true}})$ against a uniform distribution, at the fitted value and pooled over the injected values. (c) Distribution of $t(A_{\text{true}})$ at the fitted value and $\chi^2_1$.

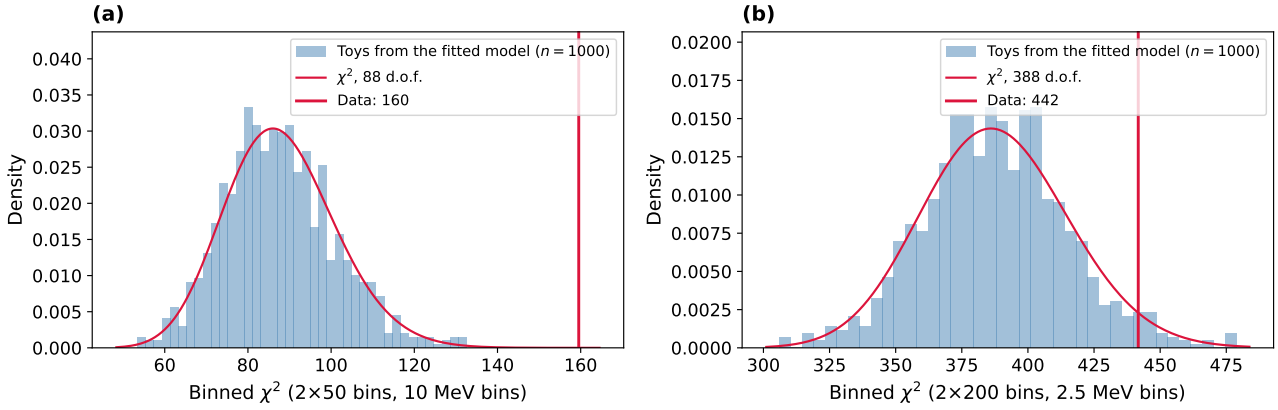



FIG. XXIV. Binned $\chi^2$ of 1000 toy-MC experiments from the fitted model, $\chi^2$ distribution with the corresponding degrees of freedom, and value of the data: (a) $2 \times 50$ bins of 10 MeV, (b) $2 \times 200$ bins of 2.5 MeV.

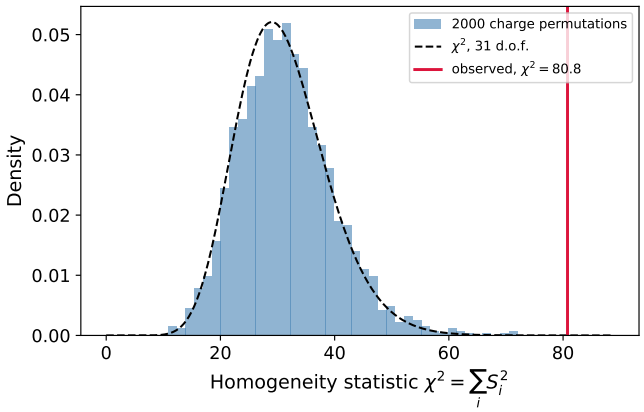



FIG. XXV. Homogeneity statistic $\chi^2 = \sum_i S_i^2$ of the $8 \times 8$ Dalitz grid for 2000 random permutations of the charges (histogram), $\chi^2$ distribution with 31 degrees of freedom (dashed) and observed value (line).

## Appendix G: Local Asymmetry in the Dalitz Plane

*In short: the local deviations and their robustness, the null tests of the procedure, the region of the published analysis and the split by magnet polarity.*

**1. Local Deviations and Their Robustness** *In short: two bins have $|S_i| > 3$ ($-3.7$ and $-3.0$) and the global homogeneity test gives $\chi^2 = 80.8$ for 31 degrees of freedom ($p = 3 \times 10^{-6}$), also in the signal window alone.*

**2. Homogeneity Test: Signal Window, Vetoes, Grids and Magnet Split** *In short: the homogeneity test holds in the signal window, with the charm vetoes and for other grids, and the structure is the same in the two magnet polarities ($\chi^2 = 23.0$ for 29 bins).*

**3. Null Tests of the Local-Asymmetry Procedure** *In short: permuting the charges gives $\chi^2 = 31.5 \pm 8.4$, as expected, and never reaches the observed 80.8; the phase-space simulation is consistent with no asymmetry, with slightly optimistic $\chi^2$ p-values.*

**4. Distribution of the Homogeneity Statistic under Charge Permutation** *In short: shuffling the charges reproduces the $\chi^2$ distribution of a uniform asymmetry, and the observed value lies far beyond all 2000 permutations.* The result is shown in Figure XXV.

**5. Asymmetry in the $\phi(1020)$ Band and in the Significant Bins** *In short: the band of the $\phi(1020)$ has no asymmetry; the negative asymmetry is outside it, and larger when the $\phi$ candidates are removed.* The result is shown in Figure XXVI.

The result is shown in Figure XXVII.

**6. Region of the Published Analysis** *In short: the $\phi(1020)$ band has no asymmetry ($-0.004 \pm 0.023$); the fit of the 2146 candidates of the published region gives $-0.234 \pm 0.025$ (Bayesian) and $-0.237 \pm 0.025$ (frequentist), against $-0.239 \pm 0.020$.*

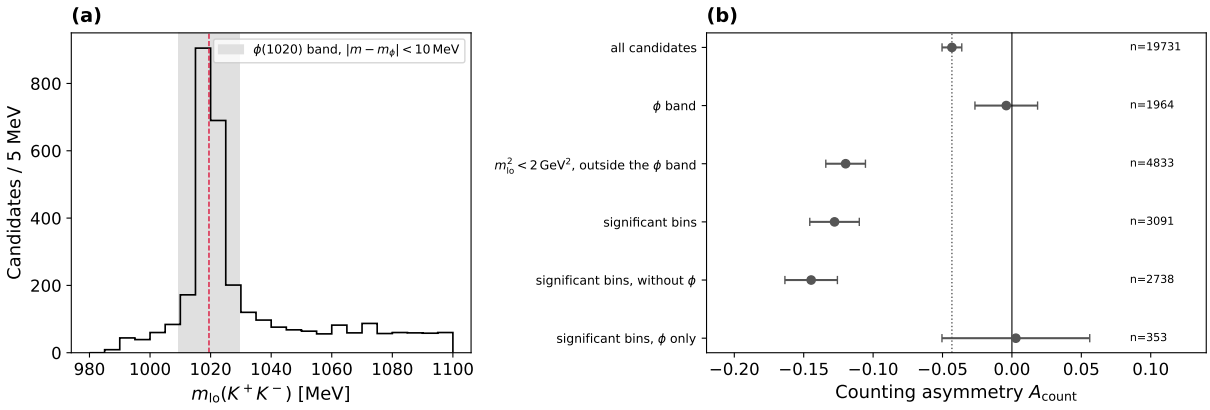



FIG. XXVI. (a) Distribution of $m_{\rm lo}(K^+K^-)$ around the $\phi(1020)$ (5 MeV bins) and band used (shaded). (b) Counting asymmetry, with binomial uncertainty and number of candidates, of all candidates (dotted line), of the $\phi$ band, of the region $m^2_{\rm lo} < 2$ GeV$^2$ outside the band, and of the two significant bins of Fig. VIII, with and without the candidates of the $\phi$ band, and of those candidates alone.

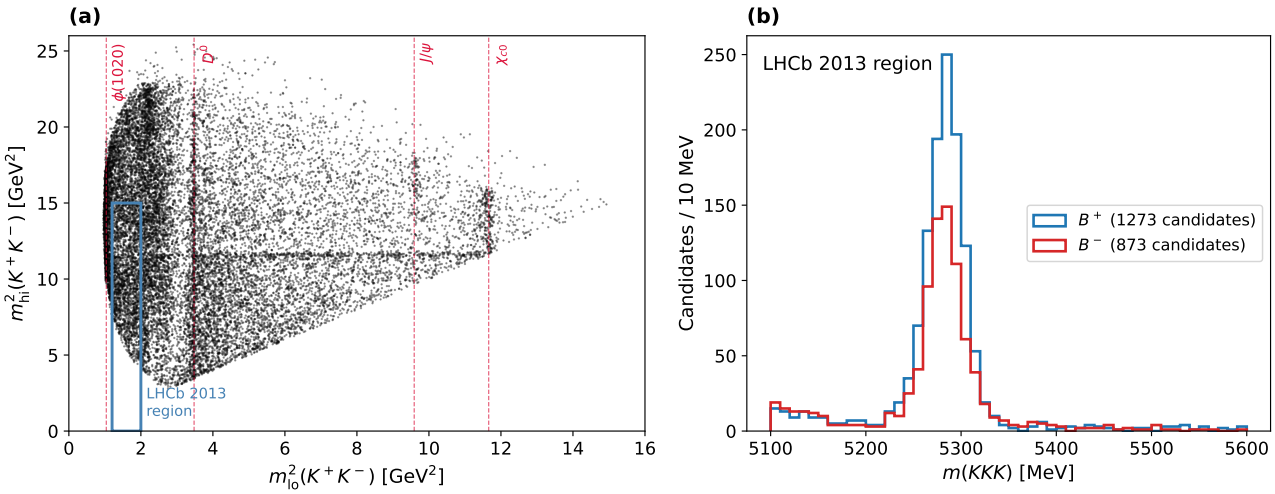



FIG. XXVII. (a) Ordered Dalitz plot of the candidates in the fit window, with the squared masses of the $\phi(1020)$, $D^0$, $J/\psi$ and $\chi_{c0}$ (dashed) and the region of the published analysis of the same data [5] (box). (b) Mass spectra of $B^+$ and $B^-$ candidates in that region, same binning: the $B^-$ peak is visibly smaller.

### Appendix H: Comparison of the Two Approaches

*In short: the numerical agreement of the two fits, the meaning of the intervals, the hypothesis tests, the profile likelihood against the posterior, and the two methods at low statistics.*

**1. Numerical Agreement** *In short: the two fits agree to the quoted digits (estimates differ by* 0.0002*, both uncertainties are* 0.0085*, systematic uncertainties* 0.0070 *and* 0.0072*); sideband subtraction gives* $-0.0503 \pm 0.0083$.

**2. Meaning of the Interval** *In short: the credible interval is a probability on A given the data and the prior, the confidence interval a property of the procedure, which the bootstrap and the toy experiments verified.*

**3. Profile Likelihood, Posterior and the Two Polarities** *In short: profiling and marginalizing the nuisance parameters give the same distribution of A; the two magnet polarities agree with each other, and combining their likelihoods reproduces the joint fit.* The result is shown in Table XII.

| | Frequentist | Bayesian |
|---|---|---|
| Magnet down | $-0.0492 \pm 0.0111$ | $-0.0497 \pm 0.0112$ |
| Magnet up | $-0.0374 \pm 0.0131$ | $-0.0378 \pm 0.0129$ |
| Combination of the two | $-0.0443 \pm 0.0084$ | $-0.0446 \pm 0.0086$ |
| Joint fit | $-0.0441 \pm 0.0085$ | $-0.0443 \pm 0.0085$ |

TABLE XII. $A_{\text{raw}}$ for the two magnet polarities and their combination, by the two methods; the frequentist combination is the sum of the profile likelihoods of the two polarities, the Bayesian one the sequential update of the posterior.

**4. Confidence and Credible Regions in** $(A, A_b)$ *In short: the* 68% *and* 95% *regions of the frequentist profile (*$\Delta$NLL *at the* $\chi^2_2$ *values* 2.30 *and* 5.99*) and of the Bayesian highest posterior density coincide, and* $A = 0$ *lies outside both.* The result is shown in Figure XXVIII.

**5. Hypothesis Test** *In short: the exact binomial test on all candidates gives* $6.1\sigma$*, the test of the fit* $p = 2 \times 10^{-7}$ *(*$5.2\sigma$*) and the Bayes factor* $BF_{10} \approx 9 \times 10^3$*; they answer different questions.*

**6. Goodness of Fit** *In short: the same fit has* $p = 0.03$ *in* 2.5 *MeV bins and* $p \approx 5 \times 10^{-6}$ *in* 10 *MeV bins: the p-value of the binned* $\chi^2$ *is a property of the binning as much as of the model.*

**7. Nuisance Parameters and Validation** *In short: the frequentist analysis profiles the eleven nuisance parameters and the Bayesian one marginalizes them; the results coincide because A is almost uncorrelated with them.*

**8. Published Analysis: Samples and Models** *In short: the published analysis has* $22{,}119 \pm 164$ *signal candidates, a Cruijff signal and misidentified and partially reconstructed backgrounds; with the same* $D^0$ *veto the Bayesian fit gives* $-0.0496 \pm 0.0088$.

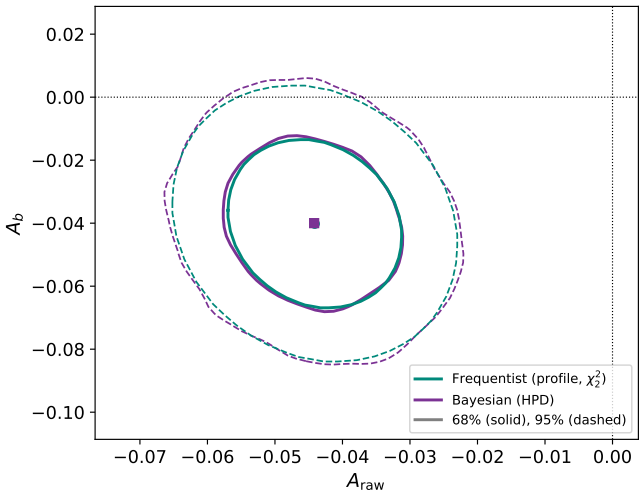



FIG. XXVIII. 68% (solid) and 95% (dashed) regions in $(A, A_b)$: frequentist, from the profile likelihood with $\chi^2_2$ thresholds, and Bayesian, highest-posterior-density regions of the chain. Dots: best-fit and posterior median.

**9. Published Analysis: Raw versus CP Asymmetry** *In short:* $A_{CP} \approx A_{\text{raw}} - A_P - A_D$*; the corrections move the published result from* $-0.060$ *to* $-0.043 \pm 0.009 \pm 0.003 \pm 0.007$ *and cannot be assumed for a different selection.*

**10. Machine-Learning Cross-Check** *In short: a cut at* 0.5 *on the classifier score keeps* 92% *of the signal and* 36% *of the background and gives* $-0.0451 \pm 0.0087$ *(Bayesian) and* $-0.0447 \pm 0.0086$ *(frequentist).*

**11. Bayesian and Frequentist Intervals at Low Statistics** *In short: credible and confidence intervals coincide for* $\sim 2 \times 10^4$ *candidates with flat priors, and stay equal, in value and coverage, down to* 50 *candidates.* The result is shown in Figure XXIX.

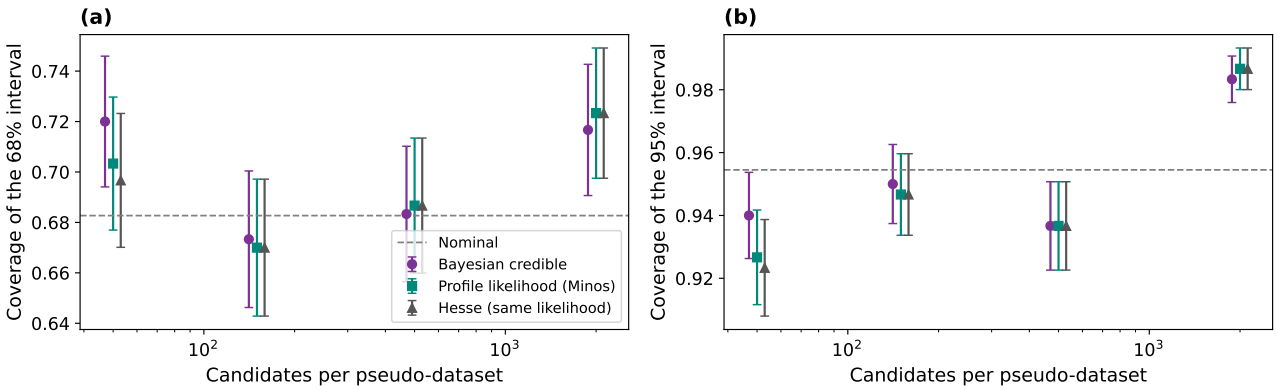



FIG. XXIX. Coverage of the (a) 68% and (b) 95% intervals on $A$ vs the number of candidates, for the Bayesian credible interval, the profile-likelihood (Minos) interval and the Hesse interval of the same likelihood; 300 toy-MC datasets per point, binomial uncertainties. The dashed lines mark the nominal coverage ($1\sigma$, $2\sigma$; the Bayesian 95% interval is the 2.5–97.5% one).

## Appendix I: Machine-Learning Analysis

*In short: classifiers trained on weak labels from the mass spectrum (there are no true labels) reach a test AUC of* 0.86*; a cut on their score leaves* $A_{\text{raw}}$ *unchanged, and an adaptive partition of the Dalitz plane shows the same pattern as the fixed binning.*

**1. Background: Regularization, Bias-Variance and Loss Functions** *In short: cross-entropy is a Bernoulli negative log-likelihood, L1 and L2 penalties shrink the coefficients, the test error splits into bias, variance and noise, and class weights rescale the score by a known factor.*

**2. Background: Classification Without Labels** *In short: the classifier that is optimal between two mixtures of the same signal and background is monotonic in the signal-to-background ratio, so labels derived from the mass are sufficient.*

**3. Background: Clustering and Uncertainty Estimators** *In short: K-means converges to a local minimum of the inertia, the adjusted Rand index measures the stability of a partition, and the spread of an ensemble estimates the epistemic uncertainty.*

**4. Weak Labels and Feature Set** *In short: a classifier separates a signal-enriched window from the mass sidebands (*15,130 *signal-like and* 1940 *background-like candidates,* 20 *features,* 60/20/20 *split); by the Neyman-Pearson argument of the CWoLa approach it is also the signal/background classifier, if the features carry no information on the mass.* The result is shown in Figure XXX.

**5. Model Comparison** *In short: six classifiers; the tree ensembles and the neural network reach test AUC* 0.857–0.864*, about* 0.09 *above the linear models and indistinguishable from each other within the test-set uncertainty (*0.012*); gradient boosting, best in cross-validation, is the reference.* The result is shown in Figure XXXI.

The result is shown in Figure XXXII.

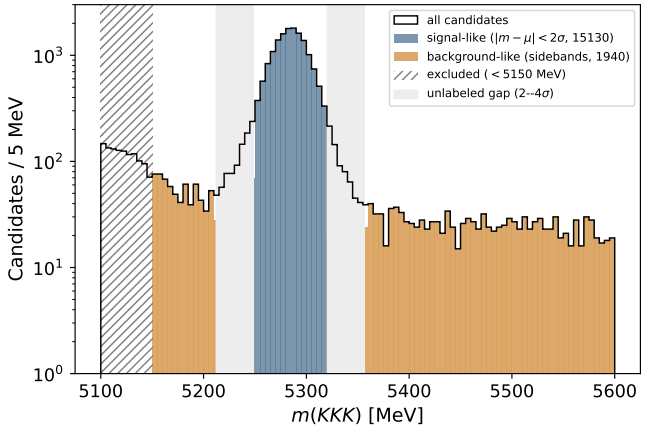



FIG. XXX. Weak labels on the mass spectrum of the fit window: signal-like candidates ($|m-\mu|<2\sigma$, 15,130) and background-like candidates (sidebands beyond $4\sigma$ and above 5150 MeV, 1940); the 2–$4\sigma$ gap is left unlabeled and the region below 5150 MeV, where the background rises towards partially reconstructed decays, is excluded.

**6. Regularization, Bias-Variance, and Calibration** *In short: the linear model does not overfit, gradient boosting overfits beyond ∼150 stages without losing test AUC, and class weights bias the probabilities, which an isotonic recalibration corrects (Brier score* 0.122 → 0.062*) without changing the ranking.* The result is shown in Figure XXXIII.

The result is shown in Figure XXXIV.

The result is shown in Figure XXXV.

**7. Uncertainty Quantification** *In short: the three estimators give mean per-candidate uncertainties of* 0.178 *(Random Forest),* 0.095 *(MC dropout) and* 0.050 *(deep ensemble); they measure different sources of variability, and only the Random Forest spread visibly decreases with the training-set size.* The result is shown in Figure XXXVI.

| Model | CV AUC | Test AUC | Brier |
|---|---|---|---|
| Logistic Regression | $0.769 \pm 0.016$ | $0.771 \pm 0.013$ | 0.197 |
| Logistic Regression (L1) | – | $0.771 \pm 0.013$ | 0.198 |
| Decision Tree | $0.816 \pm 0.011$ | $0.824 \pm 0.013$ | 0.133 |
| Random Forest | $0.863 \pm 0.017$ | $0.860 \pm 0.012$ | 0.122 |
| Gradient Boosting | $\mathbf{0.866 \pm 0.016}$ | $0.864 \pm 0.012$ | 0.116 |
| FFNN | – | $0.857 \pm 0.012$ | 0.134 |

TABLE XIII. Cross-validated AUC (mean and standard deviation over the folds), test-set AUC (with bootstrap uncertainty) and test-set Brier score of the raw scores. Gradient boosting, best in cross-validation, is the reference model. The FFNN is validated on the validation set instead of by cross-validation.

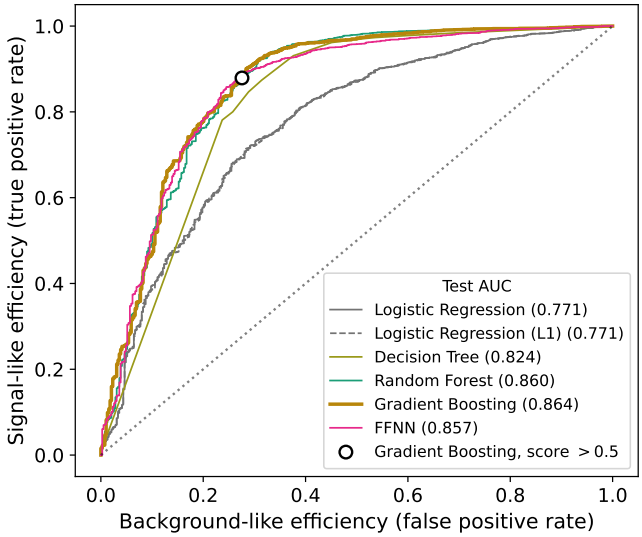



FIG. XXXI. ROC curves on the test set, with the test AUC of each model in the legend (Table XIII); the reference model is drawn thicker, and the two logistic regressions overlap. The circle marks the cut on the reference score used in Appendix I.9.

**8. K-means on the Dalitz Plot** *In short: for $K = 8$ two clusters deviate by more than $3\sigma$ ($\chi^2 = 46.4$ for 7 degrees of freedom): a negative asymmetry at low $m^2_{\rm lo}$ and positive ones at high $m^2_{\rm hi}$, as in the fixed binning.* The result is shown in Figure XXXVII.

The result is shown in Figure XXXVIII.

**9. Stability of $A_{\rm raw}$ under ML Selection** *In short: a cut on the score at 0.5 keeps 92% of the signal and 36% of the background and leaves $A_{\rm raw}$ within its expected statistical spread with both fits and with counting; it does not improve the precision and cannot remove the misidentified background.* The result is shown in Figure XXXIX.

The result is shown in Figure XL.

The result is shown in Figure XLI.

The result is shown in Figure XLII.

**10. Hyperparameter Optimization** *In short: the best configurations improve the test AUC by at most 0.005, less than its uncertainty, and the cross-validated maximum overestimates the test AUC (winner's curse).* The result is shown in Figure XLIII.

The result is shown in Table XV.

The result is shown in Table XVI.

**11. Machine Learning and the Fits** *In short: the machine-learning selection is a cross-check: a cut at 0.5 leaves $A_{\rm raw}$ within its expected spread with both methods, does not improve the precision, and cannot remove the misidentified decays.*

**12. Local Asymmetries: Fixed Binning and $K$-Means** *In short: the fixed grid and the $K$-means partition show the same pattern, a negative asymmetry at low $m^2_{\rm lo}$ and positive ones at high $m^2_{\rm hi}$ ($\chi^2 = 80.8/31$ and $46.4/7$).*

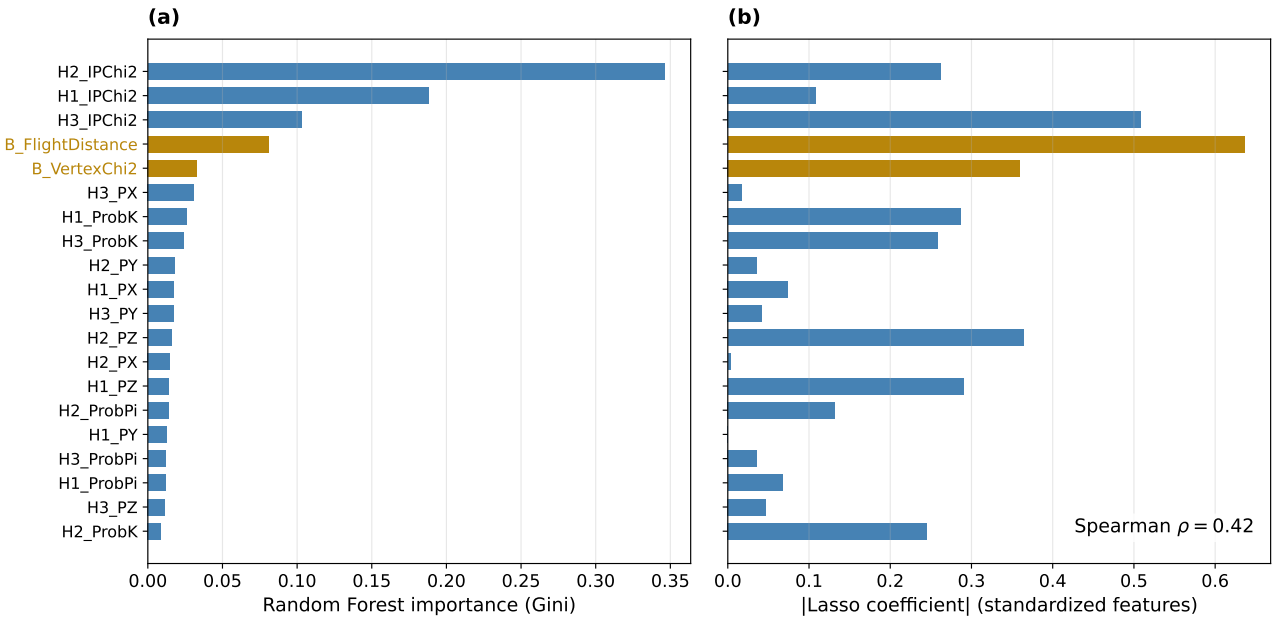



FIG. XXXII. Random Forest feature importance (Gini, a) and magnitude of the Lasso coefficient of the standardized feature (b) for the 20 features, both ordered by Random Forest importance; the two $B$-candidate variables are highlighted.

| Method | Full sample | Score $> 0.5$ | Shift (expected) |
|---|---|---|---|
| Bayesian fit | $-0.0443 \pm 0.0085$ | $-0.0451 \pm 0.0087$ | $-0.0008\,(0.0018)$ |
| Frequentist fit | $-0.0441 \pm 0.0085$ | $-0.0447 \pm 0.0086$ | $-0.0006\,(0.0012)$ |
| Counting | $-0.0432 \pm 0.0071$ | $-0.0446 \pm 0.0080$ | $-0.0014\,(0.0036)$ |

TABLE XIV. $A_{\mathrm{raw}}$ on the full fit window (19,731 candidates) and after the cut on the gradient-boosting score (15,709), by the three methods. The expected shift is the statistical spread of a nested subsample, $\sqrt{\sigma^2_{\mathrm{sub}} - \sigma^2_{\mathrm{full}}}$. Counting includes the background.

| Model | Grid | Best configuration | CV AUC / Test AUC (default) |
|---|---|---|---|
| Random Forest | `n_estimators` 200–1200, `max_depth` 6–12 or none, `min_samples_leaf` 5–40 | 400, 12, 5 | 0.8706 / 0.8656 (0.8604) |
| Gradient Boosting | `n_estimators` 100–500, `max_depth` 1–5, `learning_rate` 0.01–0.1 | 300, 2, 0.1 | 0.8684 / 0.8650 (0.8643) |

TABLE XV. Hyperparameter search for the tree ensembles (120 and 100 configurations, 5-fold cross-validation repeated 10 times, balanced class weights). The test AUC of the best configuration is compared with that of the default of Table XIII (in parentheses); the bootstrap uncertainty of each test AUC is 0.012.

| Architecture | Layers | Dropout | Learning rate | CV AUC |
|---|---|---|---|---|
| `deep_3layer` (best) | $[64, 32, 16]$ | 0.2 | 0.003 | $0.8585 \pm 0.0174$ |
| `wide_2layer` | $[64, 32]$ | 0.3 | 0.003 | $0.8582 \pm 0.0175$ |
| `deep_3layer` | $[64, 32, 16]$ | 0.3 | 0.003 | $0.8580 \pm 0.0187$ |
| `wide_2layer` | $[64, 32]$ | 0.3 | 0.005 | $0.8579 \pm 0.0184$ |
| `wide_2layer` | $[64, 32]$ | 0.4 | 0.005 | $0.8579 \pm 0.0187$ |

TABLE XVI. Top five of the 168 FFNN configurations (7 architectures from one hidden layer of 8 units to five layers $[128, 64, 32, 16, 8]$, dropout 0.1–0.4, learning rates 0.0002–0.005; 5-fold cross-validation repeated twice). The five are within 0.0007 of each other, far below the fold-to-fold spread: the optimum is a plateau. The best configuration has test AUC 0.8606, against 0.8568 for the default architecture ($\pm 0.005$ over initializations).

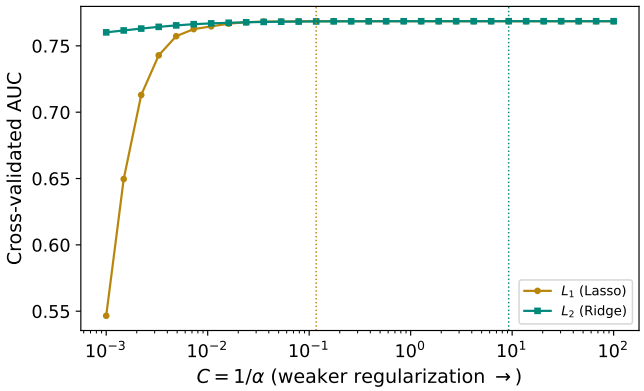



FIG. XXXIII. Cross-validated AUC of the logistic regression against the inverse regularization strength $C = 1/\alpha$ (30 log-spaced values), for the $L_1$ (Lasso) and $L_2$ (Ridge) penalties; the dotted lines mark the optimum of each.

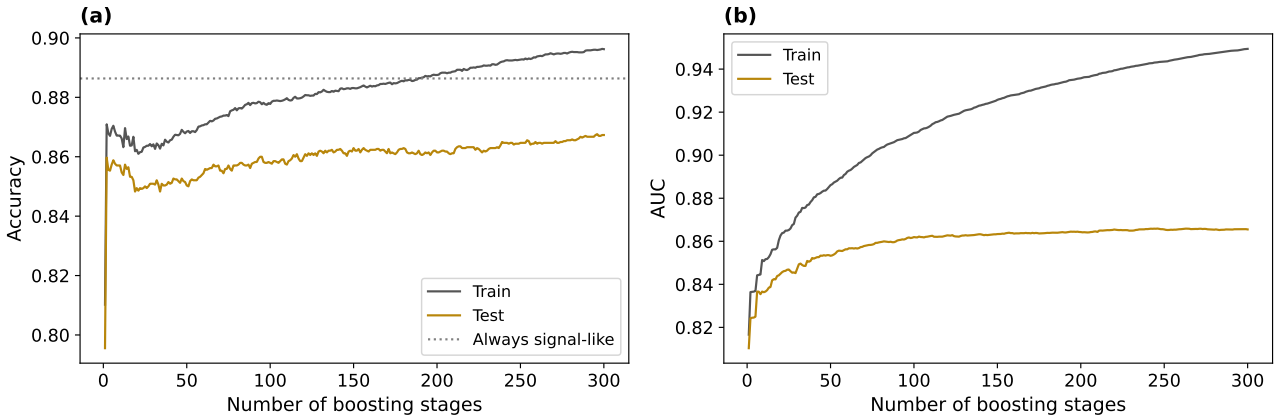



FIG. XXXIV. Gradient boosting: (a) accuracy at the 0.5 threshold and (b) AUC on the training and test sets as a function of the number of boosting stages. The dotted line is the accuracy of always predicting the signal-like class.

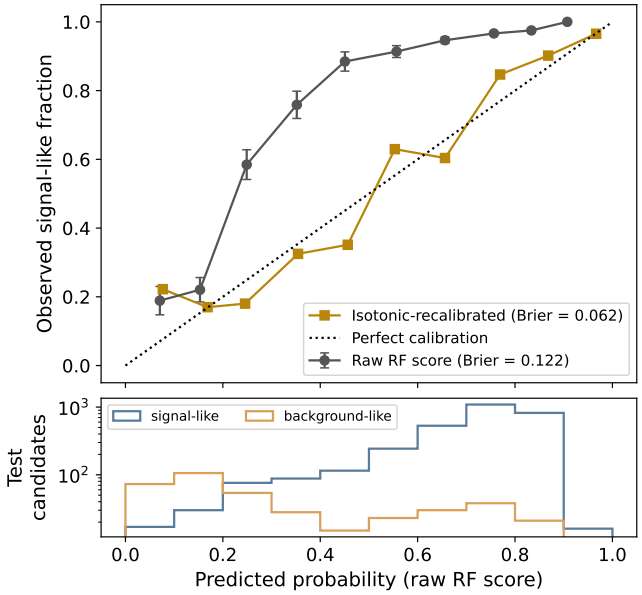



FIG. XXXV. Reliability diagram of the Random Forest on the test set: observed fraction of signal-like candidates vs. predicted probability, for the raw score (with binomial uncertainties) and for the isotonic-recalibrated one. Bottom: distribution of the raw score for the two weak-label classes; the low-probability bins contain few candidates.

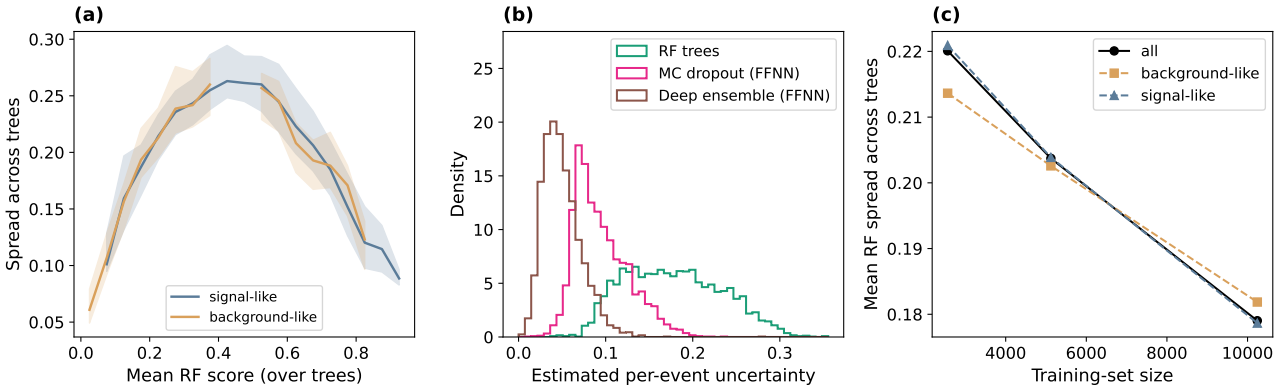



FIG. XXXVI. (a) Spread across the Random Forest trees vs. mean tree score, median and 16–84% band of the test candidates of each weak-label class. (b) Distributions of the per-candidate uncertainty from the Random Forest trees, MC dropout and the deep ensemble. (c) Mean Random Forest spread vs. training-set size, for all, background-like and signal-like test candidates.

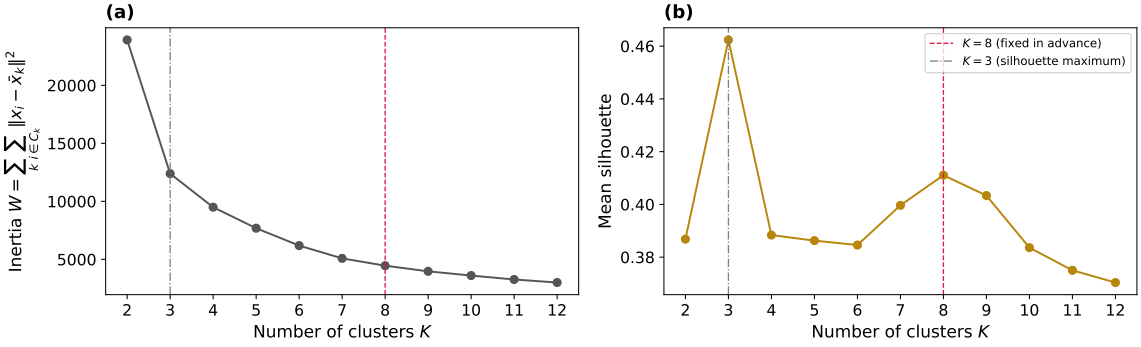



FIG. XXXVII. (a) Inertia and (b) mean silhouette of the $K$-means partition of the Dalitz plane for $K = 2$–12; the dashed line marks the $K = 8$ fixed in advance and the dash-dotted line the maximum of the silhouette, $K = 3$.

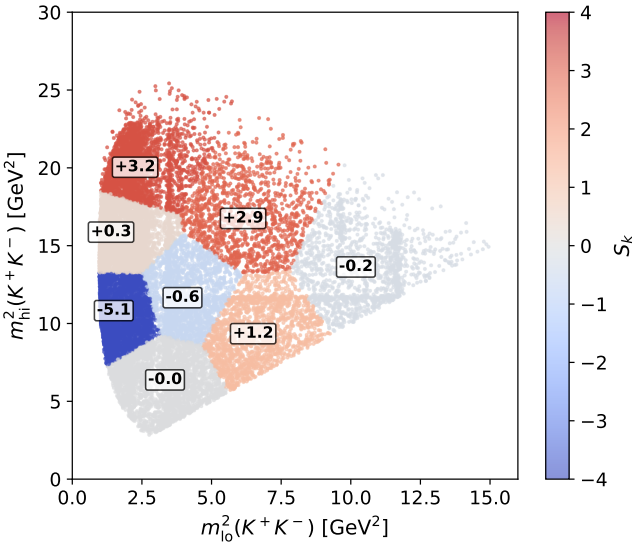



FIG. XXXVIII. $K$-means partition ($K = 8$) of the Dalitz plane, all candidates of the fit window, same axes as Fig. VIII. Color and labels: local significance of each cluster with respect to the global asymmetry, $S_k = (A_k - A_{\rm glob})/\sigma(A_k)$, on the same scale as panel (c) of Fig. VIII.

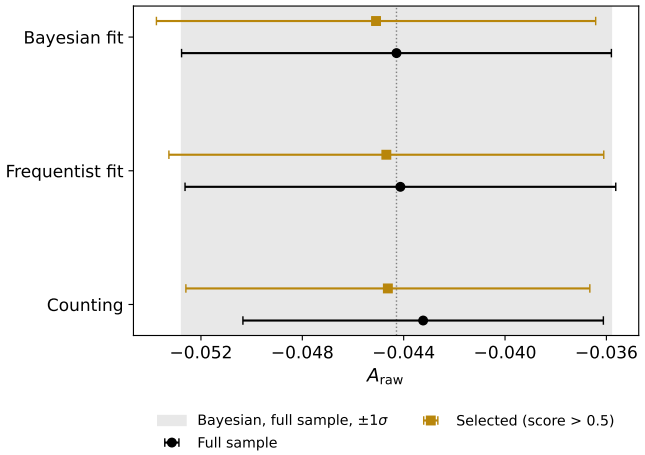



FIG. XL. $A_{\rm raw}$ from the Bayesian fit, the frequentist fit and counting, for the full sample and the ML-selected one (Table XIV).

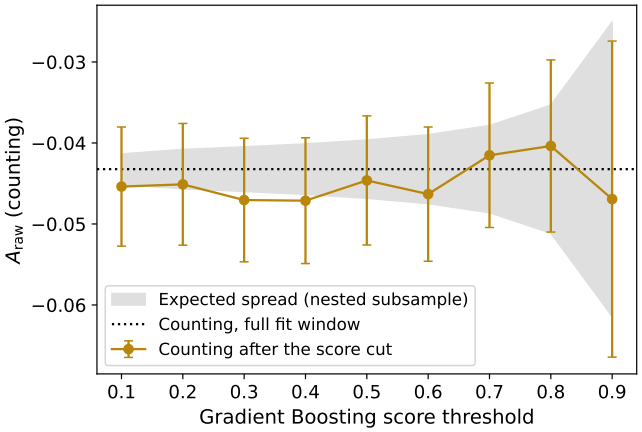



FIG. XXXIX. Counting asymmetry after a cut on the gradient-boosting score, vs. threshold, with its binomial uncertainty. Dotted line: counting asymmetry of the full fit window; band: expected statistical spread of a nested subsample around it, $\pm\sqrt{\sigma^2_{\rm sub} - \sigma^2_{\rm full}}$.

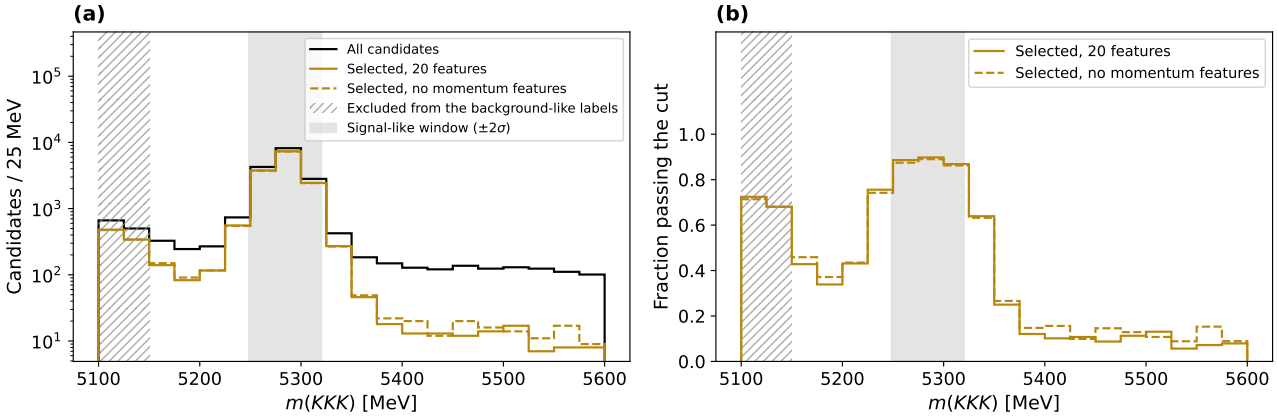



FIG. XLI. (a) Mass spectrum of all candidates and of those with gradient-boosting score $> 0.5$, for the 20-feature classifier and for one trained without the kaon momentum components (25 MeV bins). (b) Fraction of candidates passing the cut. Hatched: the region below 5150 MeV excluded from the background-like labels; grey: the signal-like window.

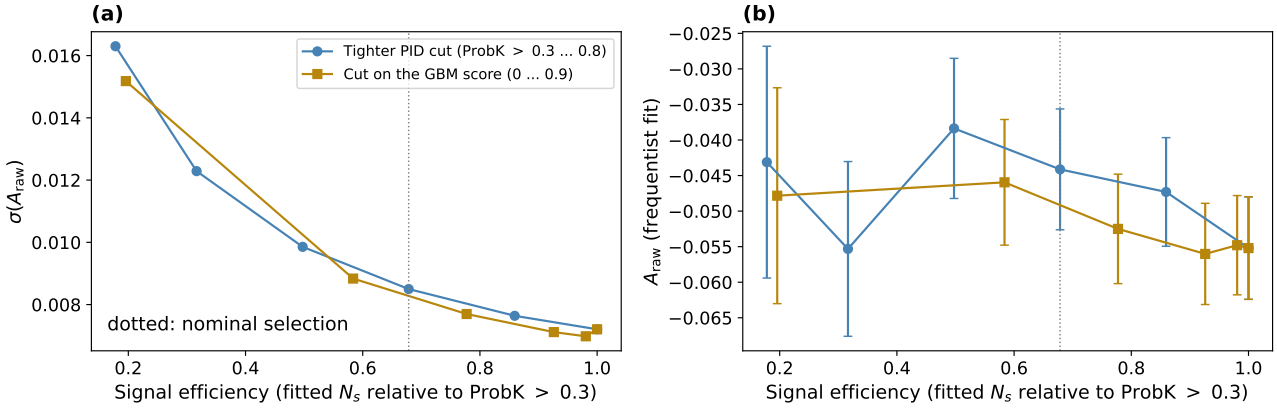



FIG. XLII. High-background sample ($\texttt{ProbK} > 0.3$): (a) statistical uncertainty and (b) value of $A_{\text{raw}}$ from the frequentist fit vs signal efficiency, for tighter identification cuts ($\texttt{ProbK} > 0.3, \ldots, 0.8$) and for cuts on the score of the classifier retrained on this sample $(0, \ldots, 0.9)$. The dotted line marks the nominal selection ($\texttt{ProbK} > 0.5$). The samples are nested, so their uncertainties are correlated.

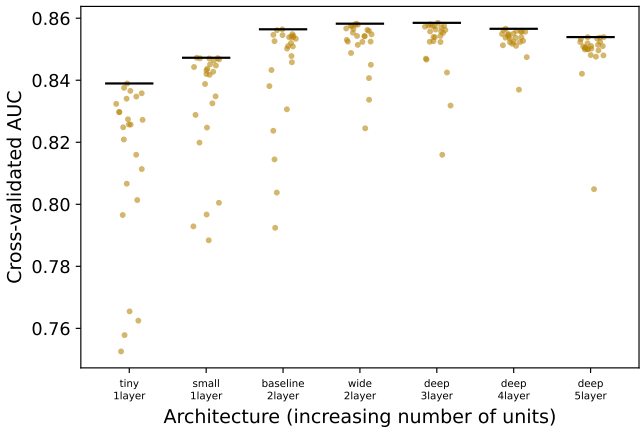



FIG. XLIII. Cross-validated AUC of the 168 FFNN configurations (dots) for each of the seven architectures, ordered by number of units; the black segment marks the best configuration of each architecture.

### Appendix J: Conclusions, Scope and Limitations

*In short: the scope of the study and the channels of the record that are not analysed, the complete text of the conclusions, and the limitations of the result.*

**1. Scope of the Study and Channels of the Record** *In short: the aim is a methodological comparison of the two frameworks on the same data, not a measurement of $A_{CP}$, which this record cannot give for lack of the control channel; the other three channels are not analysed.*

**2. Headline Result** *In short:* $A_{\rm raw} = -0.0441 \pm 0.0085 \pm 0.0072$ *(frequentist) and* $-0.0443 \pm 0.0085 \pm 0.0070$ *(Bayesian),* 5.2 *standard deviations from zero, stable under the thirteen variants; it is a raw asymmetry, not a significance of CP violation.*

**3. Lessons of Each Method** *In short: the frequentist fit gives intervals with nominal coverage and an unbiased estimator; the Bayesian one the same interval with flat priors plus diagnostics to check, a Bayes factor that depends on the prior and a predictive check; the ML selection leaves $A_{\rm raw}$ unchanged.*

**4. Systematic Study and Dependence on Choices** *In short: the particle-identification requirement dominates, and several statistics depend on choices: the binned $\chi^2$ on the binning, the Bayes factor on the prior, the coverage on how the toy experiments are generated.*

**5. Limitations and Future Work** *In short: no signal model reaches $\chi^2/\mathrm{ndof} \approx 1$ although $A_{\rm raw}$ does not depend on it, the machine-learning labels are weak and imbalanced, $A_{\rm raw}$ is not corrected for production and detection asymmetries, and the natural next step is a control-channel correction.*

### Appendix K: Code and Data Availability

*In short: Python code with the package versions listed here; frequentist, Bayesian and machine-learning code and notebooks are in the accompanying repository.*

### Appendix L: Notation

*In short: the symbols and abbreviations used in the report.* The symbols used in the text are listed in Table XVII.

| Symbol | Meaning |
|---|---|
| $A_{\rm raw}$ | raw charge asymmetry $(N^- - N^+)/(N^- + N^+)$ of the signal, parameter $A$ of the fit |
| $A_{CP}$, $A_P$, $A_D$ | CP asymmetry; production asymmetry of $B^\pm$; detection asymmetry |
| $A_{\rm count}$ | counting asymmetry of all candidates of a sample |
| $A_b$ | asymmetry of the background yields |
| $N_s$, $N_b$ | signal and background yields (both charges) |
| $\mu$, $\sigma$ | mean and width of the Gaussian signal |
| $\lambda^\pm_{1,2}$, $f_\pm$ | slopes and fraction of the double-exponential background, per charge |
| $\theta$ | vector of the twelve parameters |
| $\mathcal{L}$, $\pi$ | likelihood and prior |
| $q(A) = 2\Delta\mathrm{NLL}$ | profile likelihood ratio of $A$; $t(A_{\rm true})$ at the injected value |
| $Z$, $p$ | significance and $p$-value of the test of $A = 0$ |
| $BF_{01}$, $BF_{10}$ | Bayes factor of $A = 0$ against $A$ free, and its inverse |
| $p_B$ | posterior predictive $p$-value |
| $\sigma_B$ | expected spread of the shift between nested samples, $\sqrt{\lvert\sigma^2_{\rm var} - \sigma^2_{\rm nom}\rvert}$ |
| $N/\tau$ | chain length in integrated autocorrelation times |
| $S_i$, $S_k$ | local significance of a Dalitz bin or cluster with respect to the global asymmetry |

TABLE XVII. Main symbols used in the text.


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